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Comprehensive Analysis of Lung Cancer Mortality and Influencing Factors in Europe: A Data-Driven Approach

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ABSTRACT

This study investigates a 3.25 million record comprehensive dataset from 27 European countries to analyze lung cancer outcomes. The dataset, sourced from Kaggle, 17 explanatory variables and a response variable denoting survival status comprise it. The primary objective was to identify important variables influencing lung cancer survival and creating predictive models to improve healthcare practices.

Exploratory Data Analysis (EDA) indicated a balanced distribution of cancer stages, smoking statuses, and treatment types with almost equal representation over categories. The survival analysis indicated a significant mortality rate, with 78.04% of patients not surviving, emphasizing the severity of lung cancer.

Predictive modeling was performed on a selection of 1 million records due to computational limitations. Five models were assessed: Random Forest, LightGBM, XGBoost, Logistic regression, and Deep Neural Networks (DNN). Random Forest emerged as the most effective model with an accuracy of 73% and an ROC AUC score of 0.81, showing great discriminative power between survival outcomes. LightGBM and XGBoost, while efficient, showed lower performance with accuracy scores of 63% and ROC AUC values of 0.68. Logistic regression achieved an accuracy of 62% and an ROC AUC of 0.67, and DNN performed the least effectively with an accuracy of 52% and ROC AUC of 0.53.

Using Random Forest model, additional study revealed that survival outcomes were significantly predicted by clinical variables including cancer stage, BMI, Cholesterol, and treatment type, achieving an accuracy of 73% and a ROC AUC of 0.80. In contrast, non-clinical factors by themselves produced poor performance with an accuracy of 51% and ROC AUC of 0.52, underscoring the importance of clinical variables in predicting lung cancer mortality.

The findings highlight the effectiveness of Random Forest for predictive modeling in lung cancer research and underscore the significant role of clinical factors in survival prediction. This study provides valuable insights for improving lung cancer treatment strategies and healthcare policies across Europe.

CHAPTER ONE

Introduction

1.1 Introduction

Cancer is a condition characterized by the uncontrolled growth and reproduction of cells in a specific part of the body. These cancerous cells can invade and destroy nearby healthy tissues, including vital organs. Lung cancer is a major cause of death worldwide and a major public health problem. According to GLOBOCAN 2020 estimates from the International Agency for Research on Cancer (IARC), lung cancer continues to be the leading cause of cancer-related deaths, accounting for an estimated 1.8 million deaths, or 18% of all cancer deaths, in 2020. [1] The main cause of lung cancer is tobacco use, which includes using pipes, cigars, and cigarettes, although it can also infect non-smokers. Additional risk factors include prior chronic lung disorders, air pollution, hereditary cancer syndromes, exposure to secondhand smoke, and occupational dangers such as asbestos, radon, and certain chemicals.

The growing burden of cancer is expected to persist in the future due to the dual pressures of population growth and an aging population. While certain forms of cancer remain untreatable and fatal, others have become chronic disease rather than acute concerns as knowledge of cancer advances.

2.74 million new cases of cancer were diagnosed in Europe in 2022, a 2.3% rise from 2020. Similarly, cancer-related deaths rose by 2.4% compared to 2020. According to estimates**, lung cancer accounts for 19.5% of all cancer fatalities in the EU.** Colorectal cancer comes in second with 12.3%, breast cancer with 7.7%, and pancreatic cancer with 7.4%. [2]

This is a comprehensive patient database, mainly focused on individuals who have been diagnosed with cancer. It is designed to facilitate the analysis of different factors that could influence the prognosis and treatment outcomes of cancer. The database includes a broad range of demographic, medical, and treatment related variables, providing detailed insights into each patient's condition and medical history including 27 countries in EU with the period of 2014-2024.

In order to predict lung cancer mortality and its contributing factors in Europe, this study developed and compared five machine learning models including Random Forest, Light gradient boosting machine (LGBM), Extreme gradient boosting machine (XGB), Logistic regression and DNN. Moreover, I also intended to compare the five models and determined the best model for predicting lung cancer mortality and influencing factors.

1.2 Aims and objective

The project intends to build strong models addressing the important business requirements connected to lung cancer mortality in Europe by means of a predictive analysis using advanced data mining techniques and machine learning algorithms. Using a large dataset the research aims to develop predictive models, specifically classification models, to examine and forecast lung cancer outcomes. The goal is to identify the key factors influencing lung cancer mortality, enabling healthcare providers and policymakers to implement targeted interventions and improve patient outcomes.

Objectives:

To analyze the demographic distribution of lung cancer cases across 27 European countries:

This aim is to investigate the variations in lung cancer incidence among several demographic categories all around Europe. Examining data from 27 European nations, the initiative seeks to identify trends depending on age, gender, socioeconomic level, and geographic region. These revelations will direct the creation of focused prevention and intervention plans and assist in the identification of people most susceptible to lung cancer.

To investigate the impact of medical history and lifestyle factors on lung cancer prognosis:

This objective is to investigate how medical history and lifestyle choices, including smoking, past lung diseases, BMI, cholesterol, hypertension, asthma, impact lung cancer outcomes for a patient. The study will evaluate hoe these elements affect the prognosis, therefore guiding individualized treatment approaches meant to raise survival rates and helping to identify high-risk patients.

To assess the effectiveness of different treatment modalities in lung cancer survival rates:

The project will assess and contrast various lung cancer treatment choices, including surgery, radiation, chemotherapy and combined. The research intends to identify the most efficient treatments for various patient profiles by means of analysis of survival rates linked with every treatment modality, therefore guiding clinical decision making and enhancing patient outcomes.

To identify potential causal relationships and predictive factors for lung cancer mortality:

This aim is to investigate possible causal links and identify main elements influencing lung cancer mortality. Using advanced machine learning techniques, the study will examine a wide range of factors, demographic, lifestyle, environmental and genetic-to identify the most robust predictors of lung cancer mortality. These results will be very vital for guiding public health plans meant to lower lung cancer mortality.

To Determine Whether Clinical or Non-Clinical Factors Mostly Influence Lung Cancer Outcomes:

This objective aims to assess, on lung cancer outcomes, the relative impact of clinical factors (BMI, cholesterol, hypertension, asthma, cirrhosis, treatment type, cancer stage) against non-clinical factors (Age, sex, family history, smoking status). By means of a comparison of these elements, the study will identify which are most important in affecting lung cancer prognosis and survival, therefore guiding public health campaigns and medical practice.

CHAPTER TWO

Literature review

Cancer is a complicated and varied disease, with occurrence patterns shaped by several underlying risk factors, including environmental and behavioral ones. Driven by fast population increase, greater life expectancy, the acceptance of hazardous lifestyles, and changes in reproductive habits, research shows that cancer rates are rising in economically transitioning nations. Smoking is a leading risk factor for lung cancer. Case-control and cohort studies verified in the middle of the 20th century a causal link between smoking and lung cancer. Many other studies later have looked at how smoking affects a range of diseases, including several kinds of cancer, heart diseases and circulatory problems.

Tobacco smoking is firmly established as the primary cause of lung cancer, with approximately 90% of cases believed to be tobacco related. Lung cancer risk and elements including daily smoking frequency. Depth of inhalation and age of smoking clearly show a dose response association. A lifetime smoker runs a 20-30 times higher lung cancer risk than non-smoker. However, the risk of lung cancer decreases over time after quitting smoking.

In Europe, approximately 9% of lung cancer deaths and 2% of all cancer related deaths are attributed to radon exposure in the home. The risk is significantly higher for smokers or recent ex-smokers compared to those who have never smoked. [3] . Most European nations have shown reduced lung cancer death among men since the late 1980s. With death rates ranging from 55.4 to 46.7 per 100,000 men (age-standardized to the world population), the total decline between 1988 and 2000 across the 25 nations of the European Union was more than 13% [4]. The Russian federation as well as central and eastern European nations, which had shockingly high lung cancer death rates in the early 1990s, showed downward trends as well [5]. However, Portugal and Romania were exceptions, where male lung cancer mortality continued to rise.

In women, lung cancer mortality rates across most of Europe remain significantly lower than those for men. However, the trends differ markedly, with steady increases observed in most countries. In Denmark, Iceland, Ireland and the UK, this increasing trend has resulted in shockingly high

death rates (20 to 25 per 100,000 across all age groups) [6]. In most other European countries, female lung cancer mortality rates remain below 10 per 100,000, which is significantly lower than the rates observed in North America [7].

Many studies have looked at social differences in general and cause specific mortality in different European nations. Low educational level was found to be an independent risk factor for lung cancer and mortality. Using data from the early 1990s, one study methodically examined variations in the degree and pattern of socioeconomic inequalities in lung cancer death among 10 European countries [8]. While in some southern European countries the socioeconomic gradient was either less clear cut or inverted, the results again revealed greater lung cancer death rates among men and women with lower education levels in Northern and Western Europe.

2.1 Tobacco use as the main Lung cancer causing agent

Birth cohort has demonstrated differences in smoking behavior; lung cancer rates generally follow this trend in line with the smoking prevalence in every cohort in the United States. Lung cancer death specifically peaked in the 1935-1940 birth cohorts among white females and in the 1925-1930 birth cohorts among white males. These fluctuations in lung cancer mortality closely match the patterns in cigarette smoking within these categories [9]. Born after the cohorts with maximum risk, men and women's birth-cohort risk of lung cancer gradually dropped until about 1950. Men's age-adjusted lung cancer death rate has been declining since 1990; while the rate among women is still rising, the speed of this increase has greatly slowed [10].

The spread of smoking within populations typically follows an epidemic-like pattern with four distinct stages. In stage 1, smoking is rare and mostly done by men and those from upper socioeconomic levels. In stage 2, smoking becomes more common; rates among men peak between 50% and 80%; incidence equalizes along socioeconomic levels or stays higher among the wealthy. Usually lagging 10-20 years after men's patterns, these trends also apply to women; smoking first gained popularity among women from higher socioeconomic levels. Men's smoking prevalence in stage 3 falls to about 40% as many, especially those from higher socioeconomic levels, give up smoking. Women peak in smoking rates (35-45%), but by the conclusion of the stage their rates start to drop as well. In stage 4, smoking prevalence keeps declining steadily for men and women

as well as progressively becoming a habit connected with lower socioeconomic categories. As so, the smoking epidemic causes a change from a positive to a negative connection between socioeconomic level and smoking [11]

International comparative studies looking at smoking prevalence by age, sex, and socioeconomic level show that nations are at different phases of the epidemic. Using data from surveys carried out around 1990 in 12 Western European countries, a previous study we undertook found positive correlations between education and smoking (indicating greater smoking prevalence in more educated groups) among both men and women of all ages in Portugal, among younger and older women in Spain, and among older women in Italy and France. These results imply that Southem European nations were either at the beginning of stage 3 (Spain, Italy, France) or still in stage 2 (Portugal) of the smoking epidemic. On the other hand, among men and women of all ages in Great Britain, Norway, Sweden and the Netherlands, we discovered negative correlations between education and smoking suggesting these nations were already in stage 4 of the smoking epidemic. While in Finland they were at the end of stage 3 or beginning of stage 4, unfavorable correlations were noted just among men and younger women in West Germany [12]

Variations in lung cancer rates over time and across different areas can be mostly explained by differences in smoking patterns. Early in the 1940s, female smoking became somewhat common in North America, Northern Europe, Australia and New Zealand; it remained rare throughout the 20th century in the developing globe and areas with strong societal norms against it, including the middle east. But spurred by growing social acceptance, widening markets, and tobacco industry targeted marketing, female smoking is only now starting to become increasingly common in many of these nations [13].

Particularly when compared to high- income countries, lung cancer mortality among women has been somewhat understudied in low- and middle-income nations. Since the tobacco pandemic among women started later than it did for men, it calls for continuous documenting as it develops. Lung cancer death among women rose in Europe, North America, Australis, New Zealand and parts of Asia during the first decade of the 2000s, according to previous studies; several countries recently achieved plateauing or declining rates, especially among younger women [14]. By contrast, rising incidences of lung cancer among women were documented throughout Latin America through 2000. Updated study is crucial to record and compare present trends in lung

cancer mortality and find possible intervention sites considering the worldwide differences in the tobacco pandemic among women. While patterns among younger women represent more recent change in smoking prevalence and help estimate the future burden of lung cancer, analyzing trends among older women can help one understand the course of the tobacco pandemic [15].

Lindsey A et al. reported that several nations still see rising incidence of lung cancer among women. Many high-income countries including Scotland, Hungary, Denmark, Canada, Iceland, the Netherlands, and the United States have these rates still high. Lung cancer death rates among older women are often steady or dropping in areas including Northern Europe, North America, and Oceania where the tobacco epidemic began earlier. Still, they keep rising in places where smoking became common later, including regions of Eastern, Southern, and Western Europe as well as South America. Especially in Hungary, rates among elderly women are rising by 5.4% yearly and have already exceeded the highest rates observed in American women. Younger women's patters provide information on current changes in risk variable and can guide general future trends. For example, lung cancer death rates among younger women started to drop in nations where the tobacco pandemic started earlier, such the United Kingdom and the United States, whereas global declines in mortality did not start until two or three decades later [16].

Early indicators of a possible reversal in the lung cancer epidemic in many nations, including those where the tobacco epidemic developed later, such Latin America and Southern and Western Europe, including the dropping lung cancer rates among younger women reported in almost all analyzed countries. Tobacco control policies and growing knowledge of the health hazards connected with smoking are probable responsible for these declines in smoking initiation and higher rates of smoking cessation by lowering the starting point of the habit [17].

2.2 Lung cancer mortality attributable to air pollution

Another major risk factor for lung cancer is air pollution. Lung cancer mortality linked to air pollution becomes an increasing public health issue, particularly as air quality deteriorates in many parts of the world. Lung cancer risk is substantially influenced by fine particulate matter (PM2.5) as well as other pollutants including nitrogen dioxide (NO2) and ozone. Lung cancer incidence has been linked to prolonged exposure to these toxins, especially in urban & industrial areas.

Countries with varying income levels clearly show different degrees of air quality. Though it is a major public health issue in developed nations, air pollution is much more severe in developing countries. In some areas, fast population increases, and extensive industry have led to areas with poor air quality, therefore seriously threatening public health. Based on cohort studies, early evidence of a positive link between air pollution and lung cancer was presented in the United States in 1993 [18]. Results of long-term exposure cohort studies were published in Europe in 2000; Dutch research indicated that air pollution may be a risk factor for lung cancer. Furthermore, a 2003 Norwegian study revealed a notably higher risk of lung cancer connected to nitrogen oxides (NOx) [19].

Lelieveld and colleagues estimated in 2010 that atmospheric fine particulate matter (PM2.5) accounted for almost 6% of all deaths worldwide, or almost 3.15 million deaths [20]. According to a more recent Global Burden of Disease (GBD) research, ambient air pollution (GBD 2015 Risk factors collaborators, 2016) was associated to 4.2 million deaths, or 7.5% of all worldwide mortality, in 2015. These studies excluded household exposure from biomass burning and just looked at outdoor PM2.5 levels. PM2.5 is thought to account for 9% of all mortality in France. Regarding cancer, the International Agency for Research on Cancer (IARC) categorized outdoor PM2.5 as carcinogenic to humans, especially in relation to lung cancer [21].

Another significant air pollution is ozone, which is prevalent in urban areas, where it affected up to 74.7% of the total EU population in 2019. Ozone exposure over long term has been linked to increased death from respiratory causes as well as major negative pregnancy effects [22].

2.3 Lung cancer mortality attributable to chemical substances

Lung cancer is more likely in those who expose themselves to different harmful chemicals, especially in some jobs. 9-15% of lung cancer cases have occupational carcinogen exposure. One prominent example that either directly or indirectly causes lung cancer by inflaming the lungs is asbestos. Every commercially sold type of asbestos raises cancer risk; the danger rises with increasing length of exposure. Asbestos exposure and smoking used together greatly increase the risk of lung cancer, well above what would be anticipated from either one cause.

Similarly, lung cancer risk is linked to radon, a naturally occurring byproduct of the radioactive elements found on earth. Radon exposure levels vary geographically, with underground miners facing the highest risk. However, even modest radon levels entering dwellings raise the resident's lung cancer risk. 3-14% of incidences of lung cancer are attributable to radon exposure.

Other occupational chemicals also raise lung cancer risk. Among these are arsenic, used in wood preservation, pesticide application, and ore smelting; ionizing radiation, especially in uranium mining; vinyl chloride in papermaking; beryllium in jewelry, ceramics, missile technology, and nuclear reactor work; chromium in stainless steel production, welding, and hide tanning; nickel in electroplating, glass working, metal working, and battery, ceramic, and jewelry manufacturing; and diesel exhaust, especially among miners [23].

2.4 Random Forest classifier

Random forest is a widely used machine learning algorithm, developed by Breiman and Adele Cutler, that aggregates the outcomes of several decision trees to get a single outcome. Its simplicity and adaptability help to explain its popularity, making it suitable for both classification and regression tasks. Usually trained in the Classification and Regression Tree (CART) technique, decision trees seek to find the best spilt to divide the data. The quality of these split is evaluated using several criteria such as mean square error (MSE) and Gini impurity. Popular supervised learning methods, decision trees have problems including overfitting and bias considering their popularity. But when aggregated inside the random forest method, they often produce more accurate forecasts, especially in cases when the individual trees are uncorrelated with one another.

Multiple classifiers such as decision trees whose predictions are aggregated to identify the most common outcome make up ensemble learning techniques. Boosting and bagging (bootstrapped aggregation) are the most often used ensemble methods. Leo Bierman first proposed the bagging technique in 1996, whereby random samples from a training set with replacement let the same data points be selected several times. An expansion of the bagging technique, the random forest algorithm combines bagging with feature randomness to produce an uncorrelated ensemble of decision trees. Feature randomness also referred to as feature bagging or the "random subspace

method", generates random subsets of features, hence lowering the tree correlations. While random forests only evaluate a subset, therefore improving their robustness and accuracy, this method differs from conventional decision trees in that they analyze all possible feature splits.

Three main hyperparameters of random forest algorithms such as node size, tree count, and feature sample count that must be chosen before training. The random forest classifier then be used on both classification and regression projects once these parameters are set. A random forest is an assembly of decision trees built using a bootstraps sample, a random selection of data taken from the training set with replacement. A random forest's many decision trees make the classifier less likely to overfit the model. Average predictions from uncorrelated trees lowers general variance and prediction error. Furthermore, random forests provide a simple approach to evaluate the model's variables significance. Techniques such as Gini significance and mean drop in impurity (MDI) allow one to assess feature importance by gauging the loss in model accuracy upon the exclusion of a particular variable.

2.5 The LightGBM (Light Gradient Boosting Machine) classifier

LightGBM, short for Light Gradient Boosting Machine, is a free and open source distributed gradient boosting framework for machine learning, originally developed by Microsoft. Designed on decision tree techniques, it finds use in many different machine learning projects like classification and raking. The structure is especially meant with an eye on scalability and performance. Usually decision trees, several weak learners are merged under this framework to create a strong predictive model. The key idea is to sequentially add trees that correct the errors made by the previous trees, thus improving the overall model performance.

Decision trees as base learners: The decision tree forms the fundamental LGBM classifier building piece. Every tree in the group concentrates on fixing the errors of its predecessors, which the gradient of the loss function reveals. (variation between the projected and actual values)

Leaf-wise tree growth: Unlike most other gradient boosting methods, LightGBM develops trees in a leaf-wise rather than a level-wise fashion. Deeper, more complicated trees result from the

algorithm's leaf-wise choice to split the leaf maximizing the reduction in loss (the one with highest gradient). This can result in better performance, particularly on imbalanced datasets.

Histogram- based decision tree learning: LightGBM employs histogram-based decision tree learning to increase efficiency. This discretizes continuous feature values into bins, so that the approach can drastically cut training's memory and compute demand. When working with big datasets including many features, this approach is helpful.

Without one-hot encoding, LightGBM may directly manage categorical features, so lowering the dimensionality and supporting the maintenance of the model's efficiency. Based on its effect on the accuracy of the model, it employs a method known as "category split finding", which finds the best approach to divide the categorical feature. Its leaf-wise tree development and histogram-based approach speed it far more than conventional gradient boosting techniques. The effective memory use of the technique enables it to manage big datasets without requiring significant compute capacity. LightGBM can scale over several machines and processors by supporting both parallel and distributed training, hence accelerating the training process.

2.6 XGBoost (eXtreme Gradient Boosting) classifier

XGBoost (eXtreme Gradient Boosting) is a highly efficient and scalable machine learning algorithm that has grown to be among the most often used tools for classification tasks, particularly in contests and practical applications. Designed for speed and performance, XGBoost is an implementation of gradient boosted decision trees developed by Tianqi Chen and colleagues. XGBoost initially designed as a terminal application accessible with a libsvm configuration file. Once applied in the winning solution of the Higgs machine learning challenge, it became somewhat well-known among the machine learning community. After this success, Python and R packages were created and XGBoost has subsequently grown to incorporate implementations for languages including Java, Scala, Julia, Perl and others. Built on the gradient boosting architecture, XGBoost forms a powerful predictive model by aggregating several weak learners (usually decision trees). The main concept is to consecutively add models to reduce the mistakes caused by earlier models, hence optimizing a loss function by gradient descent.

XGBoost is the fastest and most resource-efficient tool. It comprises algorithmic improvements like parallel processing, out-of-core computing for big datasets, and cache-aware access pattem utilization. XGBoost stands out among other gradient boosting methods mostly in part from its built-in regularization to prevent overfitting. In the loss function, XGBoost combines L1 (lasso) and L2 (ridge) regularization terms to help to lower model complexity and enhance generalization.

XGBoost is highly flexible and effective, handling missing data naturally by choosing the best direction for every missing value during training, hence lowering the requirement for preprocessing tasks including imputation. With limited datasets, it is especially successful at managing high percentages of missing or zero values often present in disciplines including natural language processing and recommendation systems. Offering significant versatility across many issue types, XGBoost offers a broad array of objective functions including regression, binary and multi-class classification, ranking and custom objectives. With support for distributed computing systems like Hadoop and Spark, XGBoost can also manage vast datasets that might surpass memory constraints and be scaled effectively using multiple cores and distributed settings.

2.7 Logistic regression

Logistic regression is a frequently used statistical technique for binary classification problems, where predicting the probability that a given input belongs to one of two classes. Though its name suggests otherwise, logistic regression is really a linear model applied in classification rather than regression. Particularly for its simplicity, interpretability and efficiency this is a basic instrument in statistics and machine learning. In logistic regression, the relationship between the input variables (independent) and the output variables (dependent) is modeled using the logistic function, also known as the sigmoid function. This ability fits for probability prediction since it transfers every real-valued number into a value between 0 and 1.

Finding the best-fitting model to explain the link between a binary dependent variable and one or more independent variables is the central goal. Usually with a threshold of 0.5, the projected probability is then applied to assign a class label.

In statistics, binary variables are widely employed to represent the probability of a particular event or class occurring, such as forecasting whether a team will win or whether a patient is healthy. The most often applied method for binary regression since about 1970 has been the logistic regression model. Binary variables can be extended to categorical variables when there are more than two possible outcomes; binary logistic regression is thus generalized to multinominal logistic regression. Ordinal logistic regression is suitable in cases when the categories are ordered. Though it does not intrinsically perform statistical categorization, the logistic regression model itself represents the probability of an outcome depending on input factors. By choosing a threshold value, one can adjust it for classification problems: inputs with projected probabilities over this threshold are categorized into one category, those below another. One often uses this approach to build a binary classifier.

Linear models for binary variables can use different sigmoid functions instead of the logistic function, with the probit model being one of the most notable alternatives. With each independent variable having a corresponding parameter, the logistic model is distinguished primarily by increasing one of the independent variables producing a multiplicative scaling of the probability of a particular event at a constant pace. This characteristic of a binary dependent variable extends the odds ratio. More generally, the logistic function is the simplest way to translate a real number into a probability since it forms the natural parameter for the Bernoulli distribution.

Usually, maximum- likelihood estimation (MLE) helps one to estimate the parameters of a logistic regression model. logistic regression lacks a closed form solution for its parameters unlike linear regression, which may be solved with a closed- form expression like ordinary least squares (OLS). Nevertheless, MLE in logistic regression performs a basic function for binary or categorical results, much as OLS performs in linear regression for scalar responses.

2.8 DNN (Deep Neural Network)

A deep neural network (DNN) is a variation on artificial neural networks (ANN). Powerful tools for a wide spectrum of machine learning activities, including classification, regression, image recognition, natural language processing and more these intermediate layers known as hidden layers, allow DNNs to represent complicated, non-linear relationship in data.

A typical DNN consists of the following layers:

Input layer: this layer receives the input data. The number of neurons in this layer corresponds to the number of features in the datasets.

Hidden layer: these layers lie between the input and output layers. A DNN typically has multiple hidden layers, with each layer containing a set of neurons. Each neuron in a hidden layer receives input from the previous layer, processes it and passes the results to the next layer. The depth of a neural network refers to the number of hidden layers it contains.

Output layer: this layer provides the final output of the network. The number of neurons in this layer depends on the type of problem. There is typically one neuron in the output layer, while in a multi-class classification problem, there are as many neurons as there are classes.

Deep neural networks (DNNs) differ mostly in depth and complexity of their structure than artificial neural networks. Usually having one or two hidden layers, artificial neural networks are somewhat shallow. Deep neural networks, on the other hand, have several hidden layers that enable their modeling of more intricate correlations in the data. For complex learning tasks, for instance, the google brain project used a neural network of millions of neurons.

Deep neural networks encompass a wide range of models, including DNNs, Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Long Short- Term Memory networks (LSTMs). Recent developments have brought attention-based networks, which concentrate on network segments to improve performance on tasks. A deep neural network's complexity rises with increasing size, more layers and neurons that calls for greater computer resources and longer training times.

DNNs are particularly well suited to GPU based architectures, which significantly reduce training costs time compared to classical CPUs. Recent developments in hardware and optimization techniques have further accelerated the training process, making it more efficient to work with large and complex networks.

Deep neural network presents various challenges and considerations. The computational intensity is one major obstacle since DNNs, particularly with big models and datasets, need considerable computer power and time. Furthermore, excellent performance with DNNs usually requires a lot

of labeled data, which could not always be accessible and hence cause a restriction. Another factor is hyperparameter tuning; DNNs contain many hyperparameters, including learning rate, number of layers, and number of neurons per layer, which must be carefully changed during a trial-and-error process, often costing a significant amount of time. Moreover, DNNs are sometimes considered as black boxes, which makes it challenging to know how they reach decisions, especially in situations where interpretability is important.

Deep neural networks (DNN) greatly advance technology in many different disciplines by finding use in a broad spectrum. DNNs are used in computer vision for picture classification, object detection, and segmentation tasks, therefore enabling machines to properly comprehend and evaluate visual input. DNNs are fundamental for tasks including sentiment analysis, machine translation, and text production in natural language processing (NLP), hence improving the capacity of systems to comprehend and produce human language. In speech recognition, DNNs are also rather important since they enable virtual assistants like Siri and Alexa to correctly process and answer spoken requests. Within the field of autonomous cars, DNNs are crucial for object identification and decision- making, so enabling the creation of self-driving vehicles. DNN s are also applied in healthcare to forecast patient outcomes, assist individualized treatment, and identify diseases from medical imaging, thereby producing more accurate and customized healthcare solutions.

Several studies have included through data including medical history, family cancer history, and living circumstances to predict lung cancer risk in the general population. Tammemagi et al. developed a model in 2011 combining elements including age, body mass index(BMI), X-ray history, education level, smoking status, smoking duration, pack years, family lung cancer history, and chronic obstructive pulmonary disease history to forecast lung cancer risk over nine year period [24]. In 2012, Hoggart et al. identified age, smoking status, age at smoking initiation, smoking duration and number of cigarettes per day as main risk factors for a one-year lung cancer risk prediction model. Unlike most previous research, the study concentrated on people between the ages of 40 to 65, a younger group for smokers, Hoggart's model attained an area under the curve (AUC) of 84% but performed poorly for non-smokers [25]. Hart et al. created a 13-risk factor lung cancer prediction model in 2018, and on the test set they obtained an AUC of 86%.

Age, BMI, heart disease, physical activity, gender, smoking status, emphysema, asthma, diabetes, hypertension, Hispanic ethnicity and race comprised the risk variables [26].

Three discriminative models such as Logistic regression (LR), Support vector machine (SVM), and Deep neural network (DNN) were used to evaluate their effectiveness in distinguishing lung cancer cases from normal cases in the study on prediction of lung cancer using epidemiological data. The results revealed that the DNN based model was the best performing predictive model within the proposed framework, achieving an average sensitivity of 76.54% and an area under the curve (AUC) of 69.24% on the test sets [27]. A study conducted in Ethiopia titles "Analysis of lung cancer risk factors" identified coughing up blood, air pollution, and obesity as the most significant risk factors for lung cancer with weights of 39%, 21% and 14% respectively out of nine assessed factors. The study also suggested a machine learning model using extreme gradient boosting (XGBoost) to identify patients' lung cancer degree of seriousness. The proposed model achieved outstanding performance with 98.9% accuracy, 99% precision and 98.9% recall on the testing dataset [28].

This work aims to evaluate several categorization techniques to obtain early detection of lung cancer. Using classification techniques like Naïve Bayes, SVM, Decision tree and Logistic regression, which achieved accuracy scores of 87.87%, 99.2%, 90% and 66.7% respectively. Based on these results, it can be concluded that SVM demonstrated the highest accuracy rate among all the classification algorithms tried on these specific datasets [29]. In this paper, several classifiers were applied to a lung cancer dataset, producing different outcomes for each. The classifiers applied included KNN, SVM, Neural Network (NN), and Logistic Regression with their corresponding accuracy rates measured. Among these, support vector machine attained 99.3% accuracy. The proposed method was applied to medical dataset, helping doctors in making more accurate decisions [30].

CHAPTER THREE

Methodology

3.1 CRISP- Data mining method

Before embarking on predictive analytics using data mining, it is crucial to recognize the complexity of the process. The success of a data mining project depends on using a strong operational approach comprising a strong process model that helps to understand and control the several interactions over the complicated process. Project failure cannot be avoided without this method.

Even those with little knowledge of data science should be able to simply follow and comprehend a consistent and repeatable method that guarantees the success of this project. The project will be guided using the cross-industry standard process for data mining (CRISP-DM) approach. Following the CRISP-DM model will let us systematically address all the important elements particular to this project, therefore guaranteeing a thorough and ordered approach. From knowledge of the business challenge to application of the final model, this approach will provide the basis for every stage of the project.

The cross-industry standard process for data mining (CRISP-DM) is an open standard framework designed for planning data mining projects. This framework has been widely adopted across various industries and has consistently proven successful in real-world applications.

The process of CRISP-DM is into:

Business understanding

Data under

Data preparation

Modelling

Evaluation

Deployment

3.1.1Business Understanding

The phase known as business understanding helps one to know the problems the company wants solved. Important chore for this phase based on the data science project management including:

Determine the business question and objective:

This work aims to use machine learning algorithms to forecast European lung cancer death. Finding important elements affecting lung cancer outcomes would enable healthcare providers and policymakers to carry out focused treatments and enhance patient care. The research is to create strong models for categorization to project lung cancer death.

Situation assessment:

Analyzing a complete cancer patient database spanning 27 European countries from 2014 to 2024 if the focus of this study. Development of a model depends critically on the availability of comprehensive demographic, medical and treatment-related statistics. Together with the computational resources needed, the project also considers possible hazards including data quality and model generalizability.

Determine the project goals

The study is to develop predictive algorithms that precisely estimate lung cancer death and pinpoint relevant causes. The model's accuracy, sensitivity, and generalizing capacity across many populations will define success. The project will also compare the performance of five machine learning models to determine the most effective one.

Project plan

The initiative will be carried out in phases: data collecting and preprocessing, model construction, model evaluation, result interpretation. Random forest, LightGBM, XGBoost, Logistic regression

and DNN are among the models slated for deployment. The plan includes strict hyperparameter tuning and validation to ensure the models' dependability and resilience.

3.1.2 Data understanding

The phase on data understanding is meant to provide a thorough awareness of the information required to solve the company issue. This phase, as outlined in data science project management, entails careful data analysis to guarantee it fits the business objectives and offers the required support for addressing the identified problem.

Collect data

For this project, the dataset comprises comprehensive patient information from 27 European countries, covering the period from 2014-2024, and has been sourced from Kaggle.

Describe data

The dataset contains 18columns including ID, age, gender, country, diagnosis date, cancer stage, beginning of treatment date, family history, smoking status, BMI, cholesterol level, hypertension, asthma, cirrhosis, other cancer, treatment type, end treatment date and survived, with a total of 3250000 rows. The target variable for prediction is the survived column. The cancer stage column includes four stages: stage 1, stage 2, stage 3, and stage 4. The smoking status column has categories like never smoked, passive smoker, current smoker and former smoker. The treatment type column lists treatment options such as combined, radiation, surgery and chemotherapy.

Explore data

Data exploration will entail pattern discovery, displaying trends, and variances between variables identification. This study will ensure that the data supports the goal of the project, that of estimating lung cancer death and identifying influential factors.

Verify data quality

The quality of the data will be assessed by checking for missing values, inconsistencies, and the appropriateness of data collection methods. Ensuring that the data meets a high standard of quality is crucial for the accuracy and reliability of the predictive models.

3.1.3 Data preparation

Data preparation comes next once one has an awareness of the facts. This phase is preparing the data for the modeling one. Data science project management defines this stage as involving several important chores:

Data selection

Data selection for this project is selecting pertinent columns and rows from a complete patient dataset comprising demographic, medical, and treatment-related information from 27 European nations spanning 2014-2024. Important factors including age, gender, country, smoking status, cancer stage, treatment type and survival status will be chosen to complement the objectives of the project, that is to forecast lung cancer mortality. Any excluded data will be justified, ensuring the selections are directly linked to the business questions and objectives.

Data cleaning

Data cleaning will be minimal since the lung cancer dataset is full of no missing values and has already been standardized. Verifying the integrity of the dataset and making sure every variable is correctly structured for analysis will take the front stage. Any necessary adjustments, such as confirming the consistency of data entries or eliminating any potential outliers, will be carefully recorded. This guarantees without requiring significant preprocessing that the dataset stays correct and ready for model use.

Feature engineering

Feature engineering in this project will not entail generating any fresh variables from the current data. Rather, the emphasis will be on properly using the given variables to maximize model

performance. Every characteristic in the dataset will be utilized exactly as they are, building and using the predictive models from their present condition.

Data integration

In this project, no external datasets have been integrated. The study will just depend on the given lung cancer dataset without combining it with other data sources such socioeconomic or environmental indicators. Predicting lung cancer mortality using the current dataset will take front stage.

Data formatting

Data formatting is crucial for maintaining consistency and usability across the dataset. In this project, categorical variables such as treatment type, cancer stage, and smoking status were converted into numerical codes, with each category encoded from 1 to 4. This conversion guarantees correct structure of the data for analysis and machine learning, therefore facilitating a more effective modeling process.

3.1.4 Modelling

The modeling phase, often the most exciting part for data enthusiasts, involves the development of machine learning models to address business questions. This step is vital in generating the final prediction models even if it might be shorter than the others. Different machine learning approaches are used and polished during this phase to create a model that most fits the corporate goals. Usually, the activities in this phase include choosing suitable algorithms, training models, tuning values, and verifying the performance of the model to guarantee it satisfies the objectives of the project.

Model selection

Random forest, LightGBM, XGBoost, Logistic regression and Deep neural networks (DNN) machine learning methods are investigated in this lung cancer mortality prediction study. These algorithms were selected because of their demonstrated ability to manage complicated

relationships inside the data and big volumes. The choosing procedure balanced innovative technologies with pragmatic relevance for healthcare decision-making by considering both the accuracy of forecasts and the interpretability of the models.

Test design

Training, testing sets were created out of the dataset to ensure robust model evaluation. Considering the huge and varied character of the data, cross validation was used to maximize model performance and lower overfitting. Reflecting real-world scenarios of lung cancer mortality across European nations, the design sought to ensure the predictive capacity of the model generalizes effectively to unseen data.

Model development

Training the chosen algorithms on the preprocessed data using the training set to fit the models constituted part of model development. During this process, hyperparameters were tuned to optimize performance. The development was also aligned with the project's objective, considering practical deployment in healthcare setting and ensured that the chosen models are not only accurate but also practical in terms of computational resources and implementation in clinical environments.

Model assessment

Technical metrics including accuracy, precision, recall, F1 score and area under the curve (AUC) were applied to evaluate the models. The models were assessed for their capacity to minimize false positives and false negatives while precisely forecasting lung cancer death. Selected based on their balance of high accuracy and generalizability, the best performing model guarantees it satisfies the corporate objective of spotting important variables affecting lung cancer mortality.

3.1.5 Evaluation

The evaluation stage focuses on the performance of the model in respect to the business objectives rather than just technical metrics. Within the framework of this lung cancer mortality prediction

project, the evaluation considers how effectively the model meets the main business objective: predicting lung cancer mortality and identifying the influencing factors to guide healthcare interventions.

Evaluate results

In the context of this lung cancer mortality prediction project, the success criteria consist in accurate death projections and identification of important influencing elements to direct medical treatments. Among the five models evaluated (Random Forest, LGBM, XGB, Logistic regression and DNN), Random Forest provided the best results with high accuracy. This model would be selected since it targets high risk groups and satisfies the business goal of raising patient outcomes. Finding the most pertinent risk factors would enable public health policies and improve individualized treatment plans, thereby lowering lung cancer death all throughout Europe.

Review process

The project process was thoroughly conducted, from data preparation to model evaluation. However, there might be areas that would profit from more research, including testing the models on bigger, more varied datasets or using feature engineering approaches. Although all phases were carried out, the next versions could investigate a more thorough examination of external elements like environmental data. Although the present outcomes are good, this project iteration provides a basis for constant development and enhancement.

Determine next steps

The results indicate that the Random Forest model is ready for application, particularly in European pilot initiatives aiming at lung cancer mortality prediction. To ensure its resilience throughout many populations, though, more research and confirmation could be required. To improve prediction accuracy, future versions might additionally include extra data sources like socioeconomic or environmental elements. If the model shows success in practical uses, it may be extended into other healthcare environments or areas.

3.1.6 Deployment

The Random Forest would be used in this lung cancer mortality prediction project to be included into a user-friendly portal reachable by legislators and healthcare professionals. Users of the model could enter patient data and get real-time forecasts on lung cancer death risk either as a web-based tool or as embedded within hospital information systems. Easy-to-understand dashboard and reports would show the outcomes, offering pragmatic analysis of important risk variables and possible solutions.

A thorough monitoring and maintenance schedule is therefore essential to ensure the model stays accurate and relevant. Regular updates would be carried out as medical guidelines change, or as fresh patient data becomes accessible. Constant observation of model performance ensures that projections stay dependable by spotting any drift or accuracy loss over time. Periodic retraining of the model and system updates to mange any new data formats or outside modifications would constitute part of maintenance chores.

3.2 Software selection and python programming language

Choosing a suitable program is a vital first step before starting any data analysis. Selecting software that satisfies all analysis criteria and meets project criteria and sub criteria is vital. Ensuring future development, competitiveness, and the general success of a project depends on software choice in great part. Software that facilitates effective data preparation, exploration, and modelling is essential for lung cancer mortality prevention project leveraging machine learning. Relevant packages, methods, and configurations should be part of the selected program to enable correct analysis and simplify data interpretation, producing clear, practical findings. Python was used in this project for data preparation automation, dataset visualization, and exploration, constructing and assessing prediction models.

Python is a flexible and powerful programming language widely applied for development, coding, data science, and analysis. Because of its large libraries and tools, which allow both structured and object-oriented programming, it is the perfect fit for data analysis. Python was employed in the data preparation stage of this lung cancer mortality prediction project to create an automated script

to mange data cleaning, transformation, and validation as well as to do statistical and visual explorations to better understand the data and its variable distributions. The procedure was conducted using several packages, such as Pandas, Matplotlib, Seaborn, NumPy and Tensor Flow.

Any study starts with data exploration and preparation since the quality of this procedure determines the accuracy of the selected model and its predictive results in great extent. Reliable outcomes depend on employing a programming language such as python, which provides effective packages for these chores.

CHAPTER FOUR

Analysis and Design

4.1 Data description

This project's study is done using a normalized CSV file comprising a thorough lung cancer patient dataset taken from 27 European countries. The dataset comprises a total of 3.25 million records, each representing an individual patient, and includes 17 explanatory variables along with one response variable, "survived", which is the objective for predictive modeling. The explanatory variables cover a broad spectrum of demographic, medical and treatment related factors, providing insightful analysis of lung cancer outcomes in many populations.

This dataset, extracted from the data science platform Kaggle (http://www.kaggle.com/datasets), is set up to enable in depth investigation of lung cancer death and its influencing variables throughout Europe. Appropriate for machine learning uses, the pre-processing and normalization of the data ensure consistency and accuracy. The abundance of data and variety of variables provide a strong basis for creating predictive models meant to find important variables affecting lung cancer survival rates and therefore enhance general healthcare policies in the area.

All the recorded dataset features together with their corresponding descriptions are listed below

Variable	**Description**
ID	Patient ID
Age	Patient age
Gender	Patient gender
Country	Patient country
Diagnosis date	Date of lung cancer diagnosis
Cancer stage	4 stages of cancer & patient belong to which stage
Beginning of treatment date	Treatment start date

Smoking status	4 smoking status (Never smoked, passive smoker, former smoker & current smoker) & patient belong to which status
BMI	Patient BMI
Cholesterol level	Patient cholesterol level
Hypertension	Patient has hypertension or not
Asthma	Patient has Asthma or not
Cirrhosis	Patient has cirrhosis or not
Other cancer	Patient has other cancer or not
Treatment type	4 treatment type (Radiation, Surgery, Chemotherapy and Combined) & patient belongs to which type
End treatment date	Treatment end date
Survived	Patient survived or not

Table 1: Detail of Variables

4.2 Exploratory Analysis

4.2.1 Analysis of Cancer stages

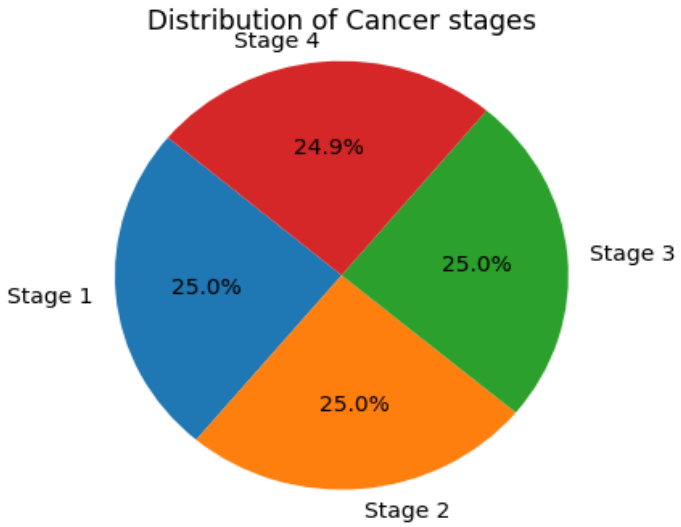

Fig 1: Distribution of cancer stages

Figure 1 states that the distribution of four lung cancer stages in dataset, each representing approximately 25% of the total cases, indicating an almost equal distribution across all stages.

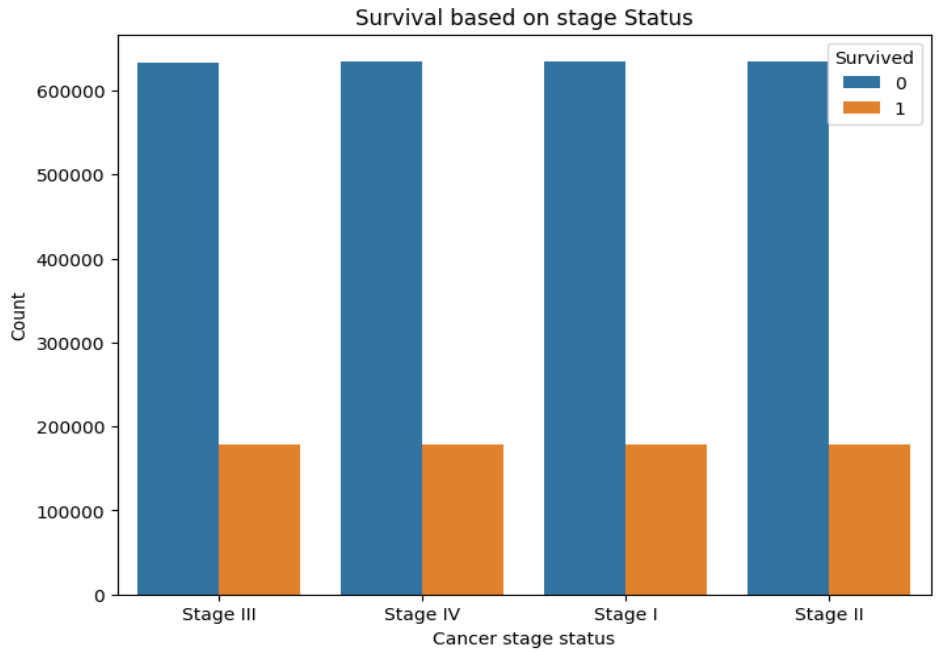

Fig 2: survival rate based on the stage of cancer

The bar chart illustrates the survival rate across the four stages of cancer (stage 1, stage 2, stage 3, stage 4) in the dataset. Reflecting the distribution of outcomes for each stage, almost 0.63 million patients did not survive while roughly 0.18 million patients survived (Fig 2).

4.2.2 Analysis of Smoking Status

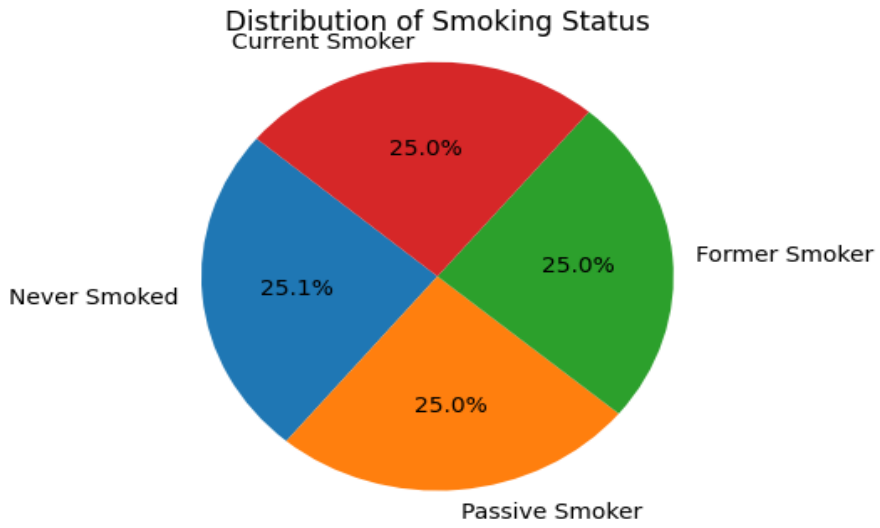

Fig 3: Distribution of smoking status

Four smoking status categories comprise the dataset: Never smoked, passive smoker, former smoker and current smoker. With each reflecting around 25% of the entire patient population, these groups are almost evenly dispersed. This balanced distribution makes it possible to investigate thoroughly how smoking behavior affects the course of lung cancer. (Fig 3)

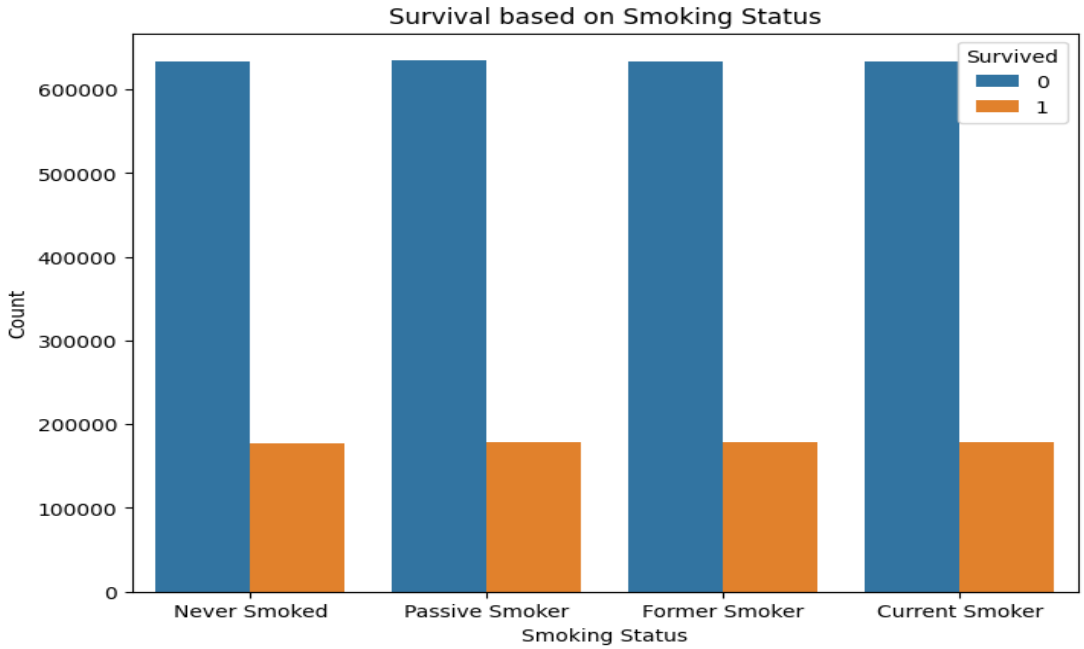

Fig 4: Survival rate based on smoking status

The bar chart (Fig 4) illustrates the survival rates across four smoking status categories (Never smoked, Passive smoker, Former smoker, and smoker) in the dataset. Reflecting the distribution of survival outcomes for every smoking status category, it reveals that around 0.63 million patients did not survive and that roughly 0.18 million patients survived. The comparison emphasizes the possible impact of smoking behaviors on lung cancer mortality.

4.2.3 Analysis of treatment type

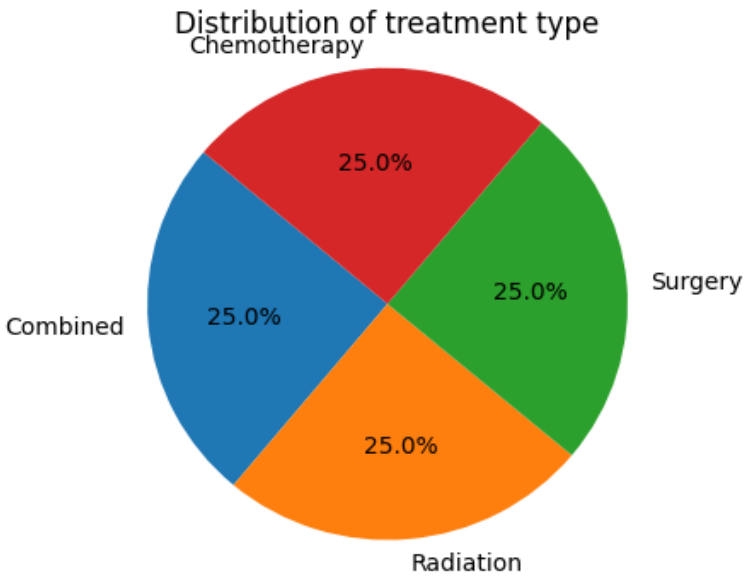

Fig 5: Distribution of treatment type

The treatment types in the dataset such as radiation, surgery, chemotherapy and combined are equally distributed, with each category accounting for approximately 25% of the total treatments. (Fig 5). This equitable distribution makes it possible to fully examine the results and efficacy of every treatment technique, therefore offering important new perspectives on their influence on lung cancer survival rate.

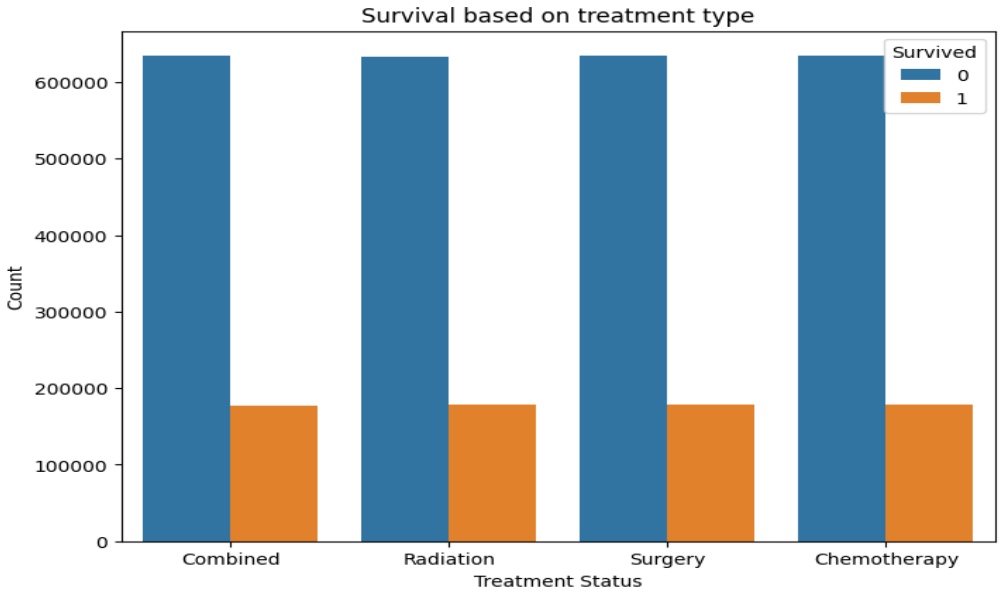

Fig 6: Survival rate based on treatment type

The survival rate over the four treatment modalities such as radiation, surgery, chemotherapy and combined are presented in the bar chart for the dataset (Fig 6). It shows the distribution of patient outcomes around 0.63 million patients did not survive and roughly 0.18 million survived. This analysis emphasizes the different survival results connected to every treatment modality and provides understanding of the general success of various interventions in cases of lung cancer.

4.2.4 Age distribution

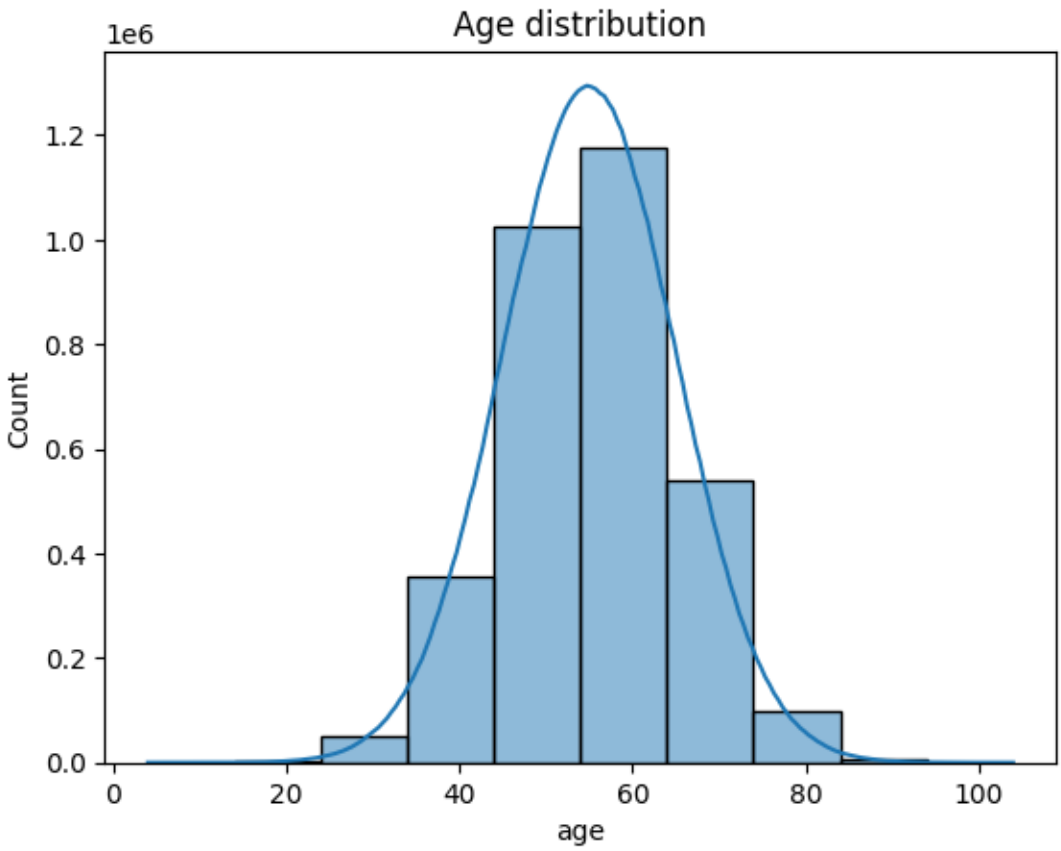

Fig 7: Histogram of age and count of lung cancer patients

In Fig 7, The age range is shown by the x-axis, and the frequency or count of lung cancer patients falling within each age range shown by the y-axis. From this histogram, the majority of patients are aged between roughly 45-65. There is an increase in frequency from age ranges around 25-50, indicating a particular proportion of patients with lower age levels. The frequency stays reasonably high between age ranges 45-65, indicating a significant proportion of patients with of patients with ages in this range. There is a decline in frequency above 65, suggesting that fewer patients have high age ranges.

4.2.5 Cholesterol distribution

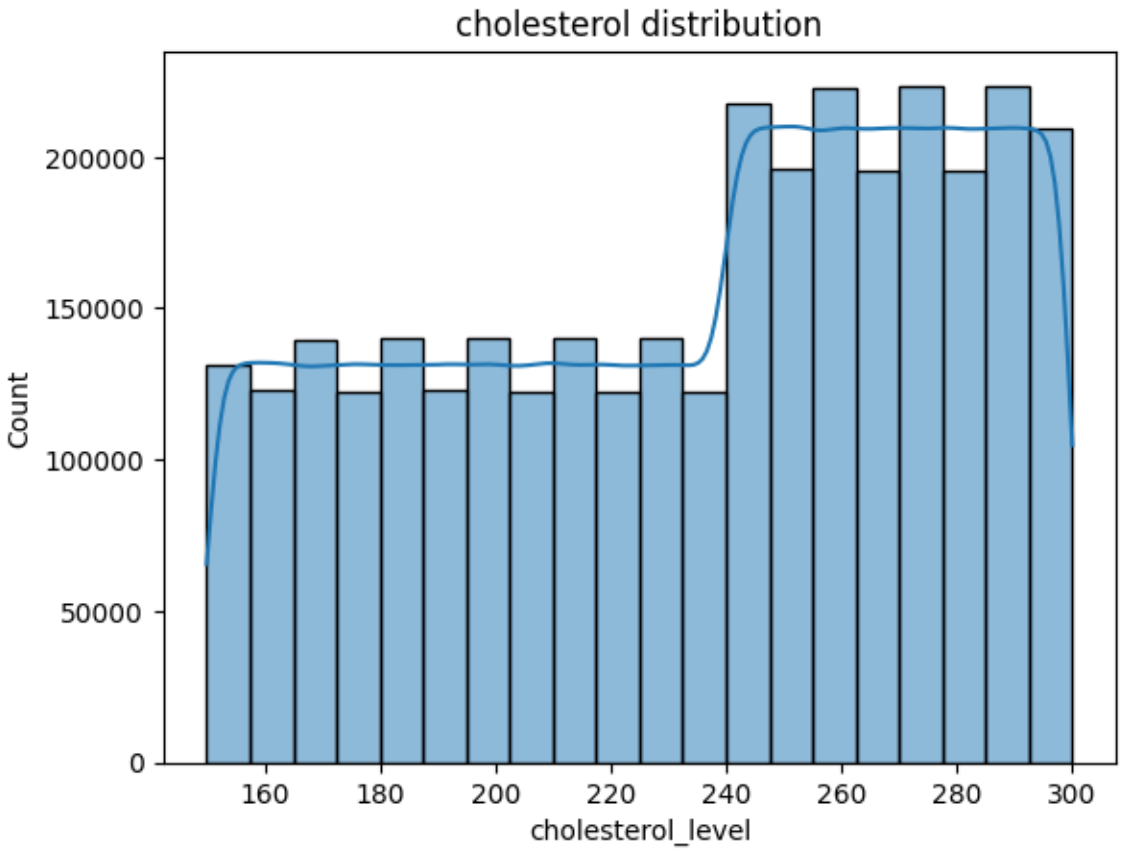

Fig 8: Bar chart of cholesterol level and count of lung cancer patients

Fig 8, The bar chart shows how cholesterol levels among the lung cancer patients are distributed. The cholesterol levels fall into various ranges, with higher concentrations observed between 240 and 300 units. There is a noticeable rise in patient count starting from cholesterol levels of 240, peaking around 280-300. This persistent trend suggests that a considerable number of patients had raised cholesterol levels, which would imply a probable relationship between high cholesterol and lung cancer death or progression. Further medical research and treatment planning could benefit from such understanding.

4.2.6 Country Distribution

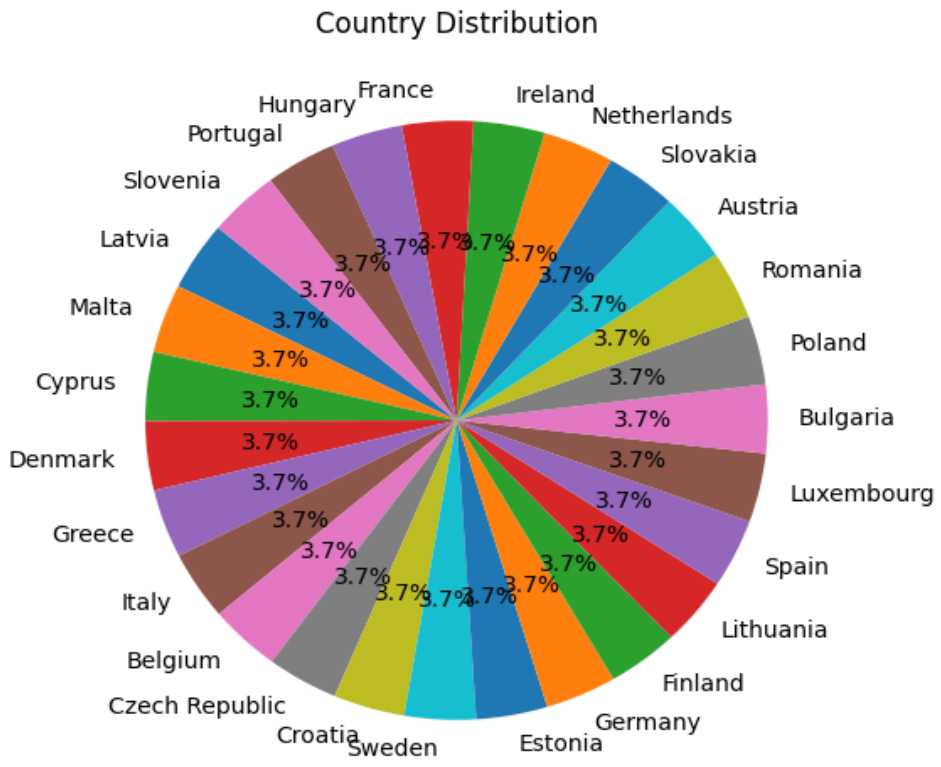

Fig 9: Distribution of lung cancer patients

Lung cancer patients are distributed among 27 European countries shown on a pie chart (Fig 9). Every country represents an equal amount of 3.7% of the whole dataset, so reflecting a balanced representation of patients from every country. Reflecting the wide geographical extent of this research, all countries equally contributed to the dataset. This consistent distribution guarantees that the study covers several communities from all throughout Europe, so offering a whole picture of lung cancer cases throughout the continent.

4.2.7 Analysis of Survival Count

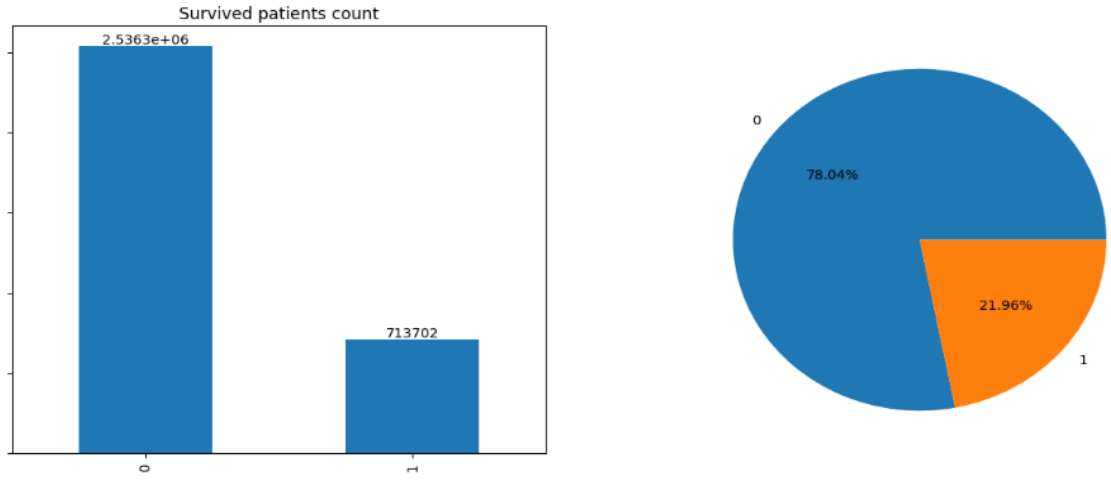

Fig 10: Total Survived patients count

Fig 10 shows how survival results in the lung cancer dataset are distributed. It shows that just 21.96% of the patients survived whereas the great majority, 78.04%, did not survive. With almost 4 out of every 5patients not surviving lung cancer, this clear contrast emphasizes the great mortality rate linked with the disease in the dataset. This survival distribution offers important new perspectives on the degree of lung cancer outcomes among the known cases.

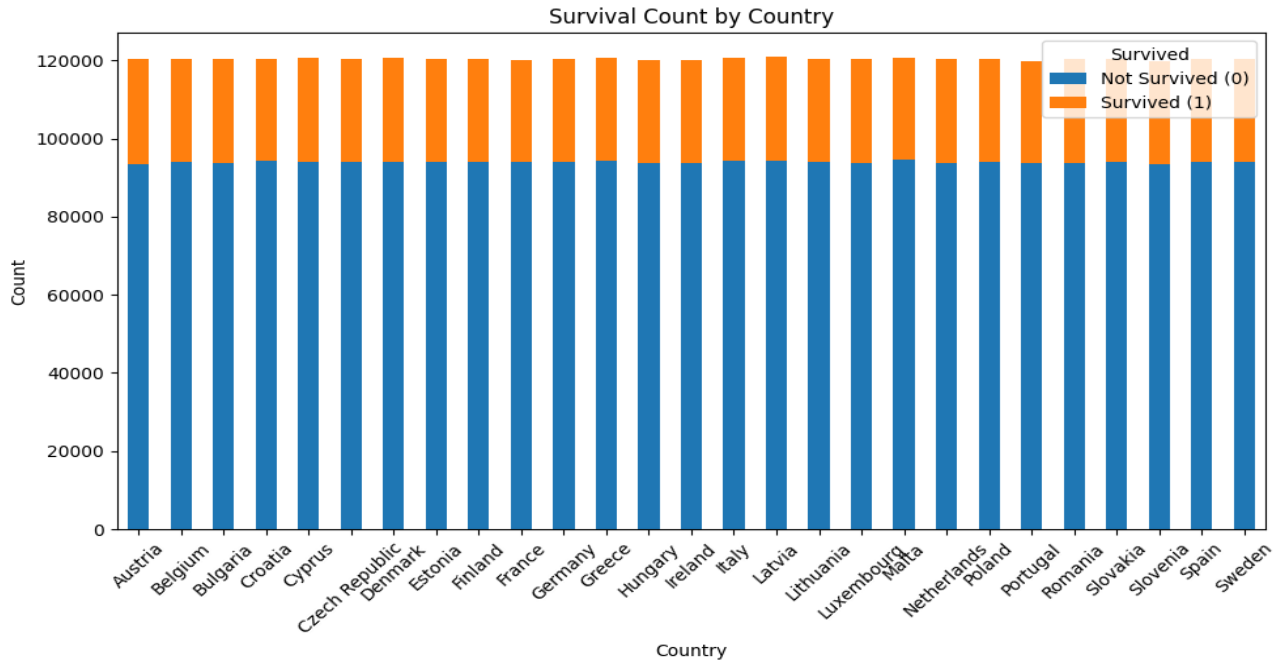

Fig 11: Survival count by country

```python
# Printing the grouped data (value count by country and survival status)
print(grouped_data)
```

```
survived            0      1
country
Austria         93474  26778
Belgium         94052  26435
Bulgaria        93663  26633
Croatia         94214  26259
Cyprus          94089  26703
Czech Republic  93953  26525
Denmark         94133  26575
Estonia         94037  26351
Finland         93880  26478
France          93887  26293
Germany         93989  26374
Greece          94294  26365
Hungary         93644  26476
Ireland         93721  26484
Italy           94223  26382
Latvia          94205  26742
Lithuania       94020  26334
Luxembourg      93844  26456
Malta           94444  26349
Netherlands     93791  26427
Poland          93951  26343
Portugal        93674  26133
Romania         93825  26452
Slovakia        93906  26327
Slovenia        93408  26229
Spain           93950  26395
Sweden          94027  26404
```

Fig 12: Survived & Not survived count by country

The survival count of lung cancer patients in 27 European countries is shown here using a bar chart (Fig 11 & Fig 12). The chart displays the number of survivors (in blue) versus those who did not survive (in orange) for each country. Split between the two survival outcomes, each bar shows the overall patient count for that country. This uniform pattern shows that, most patients in the dataset did not survive lung cancer, regardless of the country.

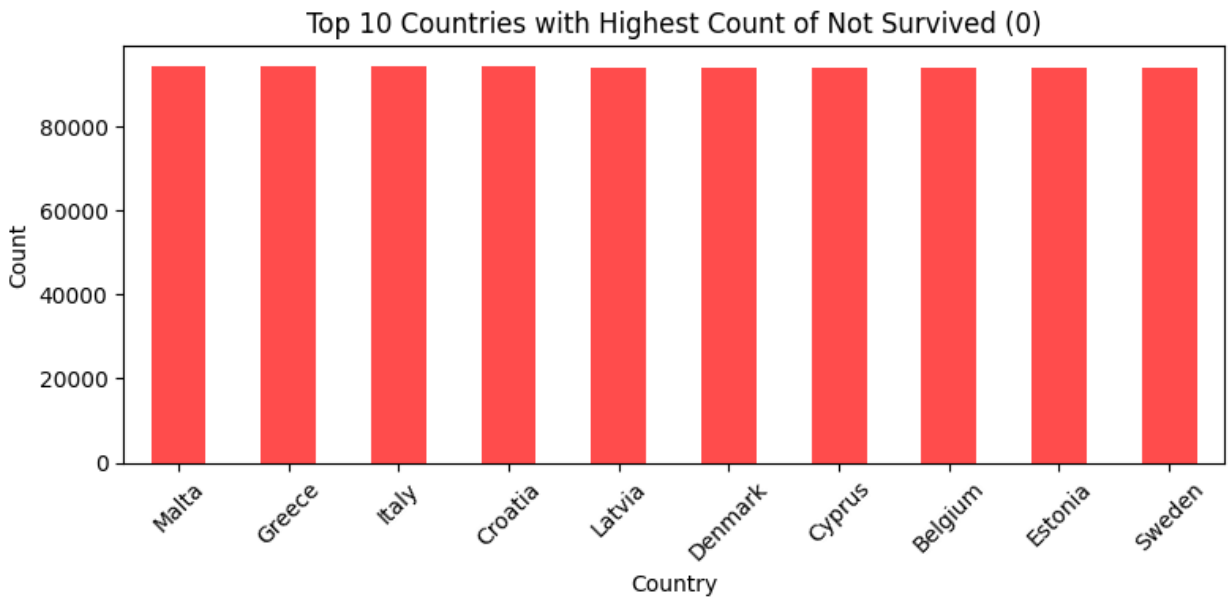

Fig 13: Highest count of Not survived

In Fig 13, the top 10 countries with the greatest count of patients who did not survive (labeled as '0') in the lung cancer dataset are shown on this bar chart. The countries, including Malta, Greece, Italy, Croatia, Latvia, and others show nearly same numbers of non-surviving patients, with each country reporting approximately 93,000 cases. The consistency of the counts across these countries points to a constantly high lung cancer death rate throughout different European Countries. The facts underline the urgent need for better healthcare policies and cancer treatment regions.

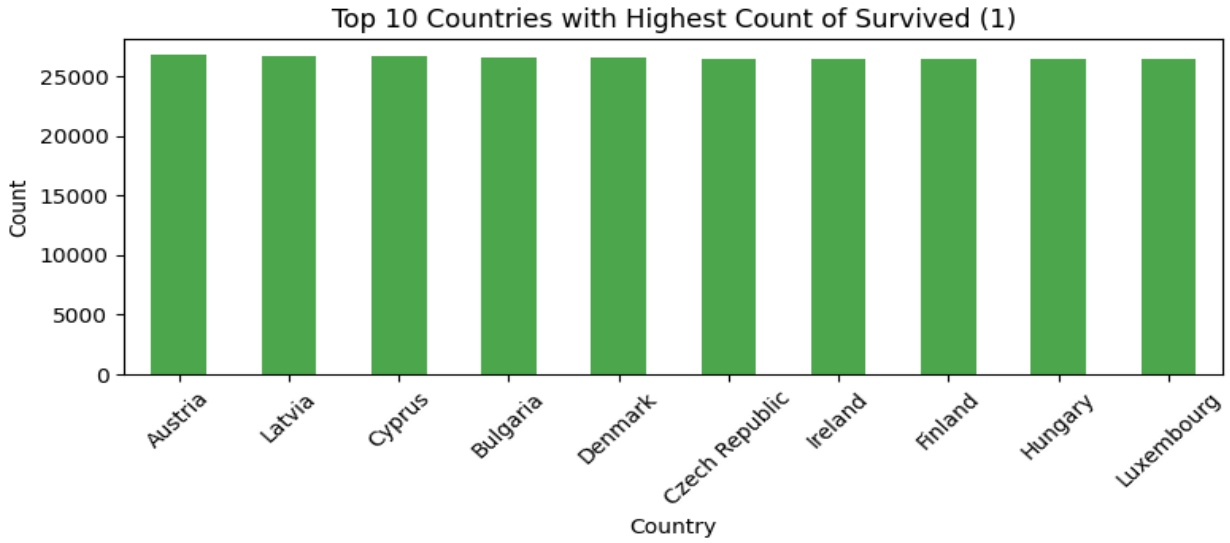

Fig 14: Highest count of Survived

Fig 14, The bar chart highlights the top 10 countries with the highest count of lung cancer survivors (labeled as '1'). Countries including Austria, Latvia, Cyprus, Bulgaria, and Denmark have 26,000 surviving patients each. The consistency in survival rates among these countries suggests that, despite the considerable death linked with lung cancer, there are rather many survivors in these countries. This information points to the need for ongoing efforts in healthcare advances and cancer treatment programs since it implies the possible efficacy of medical intervention or treatment in these regions.

CHAPTER FIVE

Implementation

5.1 Predictive modelling

The dataset used in this study originally comprising 3,250,000 records, which were extensively examined throughout the exploratory data analysis (EDA) phase. This large dataset made it possible to understand the features, distributions and possible correlations between variables completely. However, when moving to the predictive modeling phase, computational restrictions forced to reduce the dataset size, especially in environment like Google Colab, where managing such a lot of data can be difficult. As a result, a subset of 1,000,000 records was selected for the modeling phase. The dataset shows a significant class imbalance, with 780,541 non-survived instances (label 0) and 219,459 survived instances (label 1). Although it allowed for more efficient processing and faster model training, this smaller dataset was still sufficiently large to preserve a representative sample for building and evaluating the models, so enabling the predictive analysis to be caried out effectively without overloading system resources.

A predictive model is a mathematical instrument for future outcome projection grounded on past data. Although there are several modeling approaches accessible, not all of them fit every dataset. Different approaches might assist in improving the model choosing process based on the analysis and data quality. It generates informed forecasts regarding new or unexplored cases by use of data analysis of trends relations. From customer behavior to medical condition classification, predictive models find application across a broad spectrum of issues. A predictive model's accuracy and dependability relies on the quality of the data, feature selection, and model performance in identifying trends within the data. Once developed, these models can be assessed with performance criteria including accuracy, precision, recall and misclassification rate, hence guiding the generalization to new data.

The dataset was first separated into training and testing sets, with 80% of the data assigned to training and 20% set for testing, before applying the prediction models. This ensures that, leaving some for-performance evaluation on unseen data, the models can be trained on most of the data.

Since the dataset is imbalanced and with few instances of outcomes, synthetic minority over sampling technique (SMOTE) was applied on the training data. SMOTE is a method for synthetic example generation for the minority class, hence balancing class distribution. This helps to avoid the model from being biased towards the majority class, therefore enhancing the capacity to forecast minority class events properly.

Cross- validation on the training set was done following SMOTE application to ensure the model's resilience and stop overfit. Cross-validation separates the training data into many folds whereby the model is trained on a subset and validated on the other data. This process is performed several times, and the average of the outcomes offers a more accurate evaluation of the performance of the model. We want to keep a balanced representation of the classes while nevertheless increasing model accuracy and generalization by combining SMOTE with cross-valuation.

Through the modeling process, the five models' parameters will be optimized, and their outcomes will be analyzed using statistical measures. Apart from other important statistical results, the models' whole performance will be assessed depending on their misclassification rates. A comparison of overfitting between the training and validation sets of each model will also be carried out.

The Random Forest model is applied first, known for its simplicity and strength in classification problems. Following this, LightGBM and XGBoost, both gradient-boosting models, are employed for their efficiency and performance in big datasets. Additionally included is logistic regression, a proven technique for binary classification situations like this. Finally, a Deep Neural Network (DNN) model is used to provide the possibility to identify intricate, non-linear relationships in the data. By comparing these five models, the most effective model was selected based on Accuracy of the model, ROC AUC value, precision, recall, F1 score and confusion matrix.

5.2 Random Forest classifier

Random forest is an ensemble learning method that constructs multiple decision trees and aggregates their outputs to improve prediction accuracy and reduce overfitting. It provides strong management of big datasets and captures complicated interactions among variables, hence being quite effective for both classification and regression applications. In this analysis, the random forest model performed effectively, achieving an accuracy of 73%, which means it correctly classified 73% of the cases. The model's ROC AUC score of 0.81 demonstrates great discriminating ability between positive and negative classes, indicating that it can effectively separate different outcomes. While the recall score of 74% shows that the model effectively spotted 74% of the real positive events, its 73% precision indicates that it accurately identified 73% of its positive predictions. Indicating the model's dependability, the F1 score of 0.73 shows a fair trade-off between precision and recall. The confusion matrix revealed that the random forest model accurately classified a significant number of true positive and true negative, making some misclassifications resulting in 175,763 false positive and 160,282 false negatives. Random forest is a well-balanced and efficient model for this dataset since it displayed good performance across all metrics

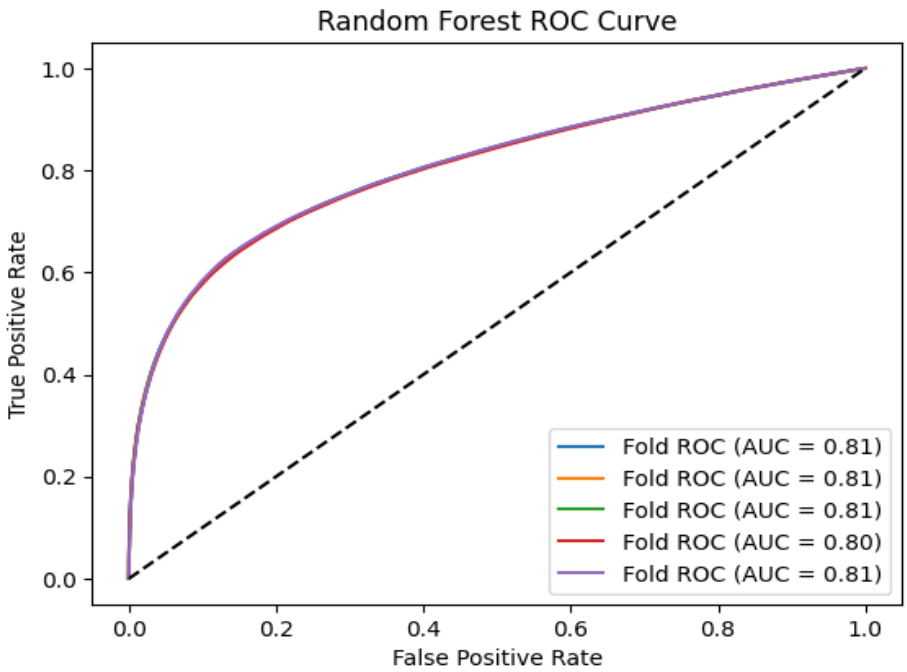

Fig 15: Random Forest ROC Curve

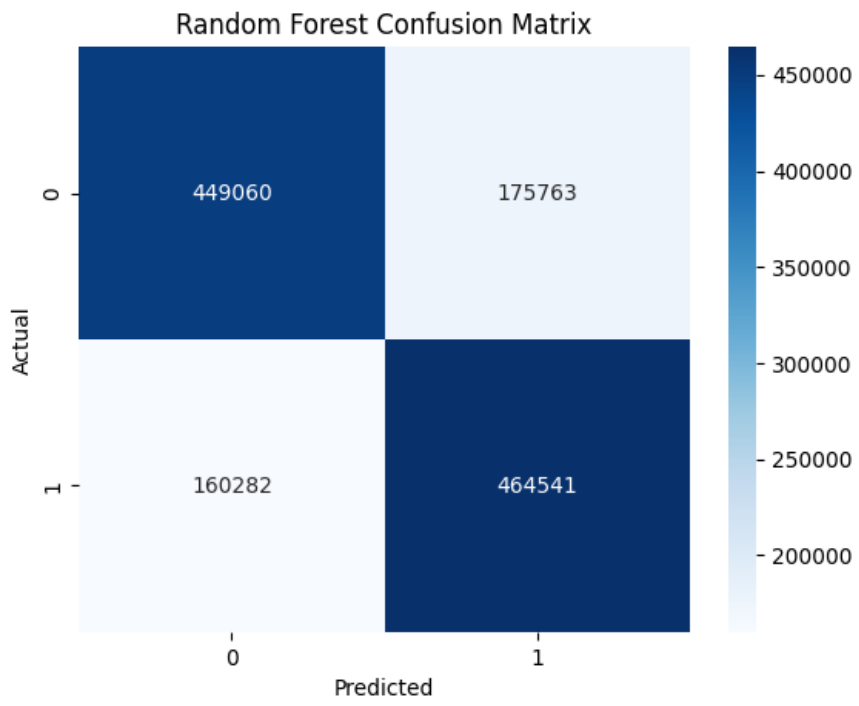

Fig 16: Random Forest Confusion Matrix

5.3 LightGBM

LightGBM is a gradient boosting algorithm specifically designed for speed and efficiency, particularly suitable for managing large datasets with numerous features. Its leaf-wise construction of decision trees enables more effective computation than traditional level-wise method. LightGBM achieved an accuracy of 63% in this study, therefore correctly classifying 63% of the cases. The ROC AUC score of 0.68, the model shows some capacity to discriminate outcomes but is not as powerful as Random Forest in differentiating positive from negative classes. Although its recall score of 63% shows that it found 63% of the real positive events, the model precisely identified 63% of its positive predictions. The F1 score of 0.63 indicates a balance between precision and recall, though generally performance is lower than that of the random forest model. The confusion matrix of LightGBM shows 232,615 false positives and 228,987 false negatives, therefore misclassifying a significant amount of data. Although LightGBM demonstrated moderate

success in handling datasets, it fell short of Random Forest in terms of accuracy or discriminative power, so it is a less ideal solution for this specific dataset even with computational efficiency.

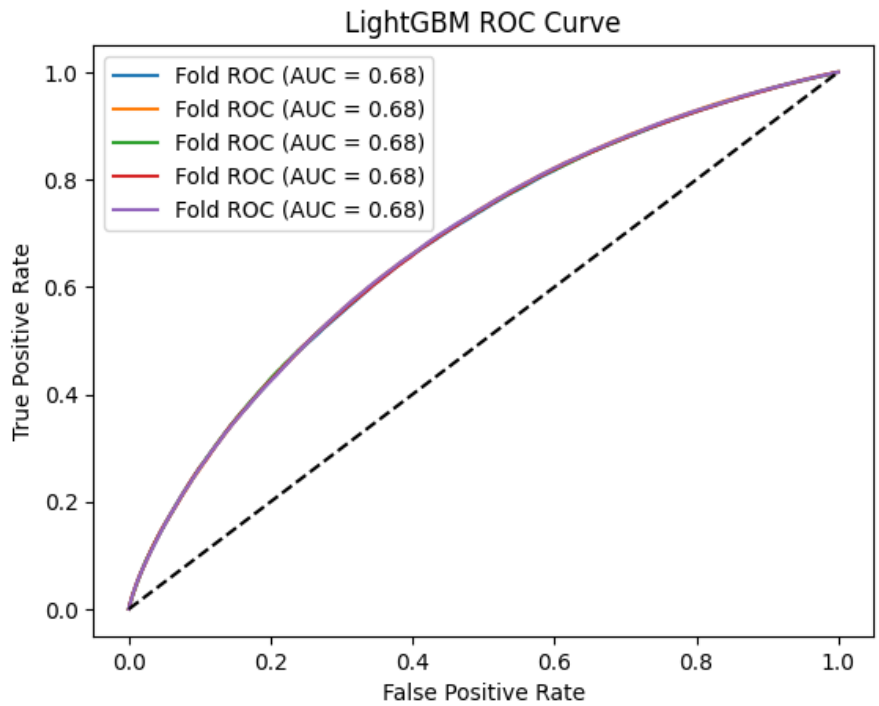

Fig 17: LightGBM ROC Curve

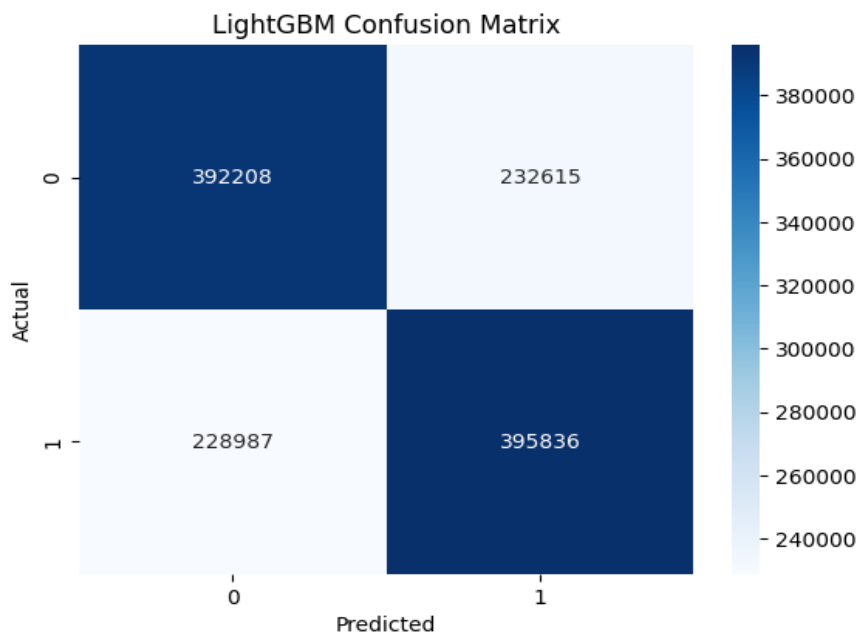

Fig 18: LightGBM Confusion Matrix

5.4 XGBoost

XGBoost is a generally utilized gradient boosting method distinguished by speed and good performance, especially useful with structured data. One of its main benefits is the use of sophisticated regularizing methods that help reduce overfitting, making it more reliable for generalization. XGBoost obtained an accuracy of 63% in this study, properly categorizing 63% of the instances. Like LightGBM, the model's ROC AUC score of 0.68 demonstrates moderate ability to separate positive from negative classes. The model successfully identified 63% of its positive predictions with a precision of 63%and has a recall score of 64% of actual positive cases, which is slightly better than LightGBM. The F1 score of 0.64 demonstrates a slight increase over LightGBM, reflecting a rather better balance between precision and recall. Though similar to LightGBM, the confusion matrix showed that XGBoost generated 231,649 false positives and 224,441 false negatives, therefore demonstrating slightly better recall. In last, while XGBoost performed similarly to LightGBM, with only a minor improvement in recall, it still did not match the higher performance levels observed in the Random Forest model.

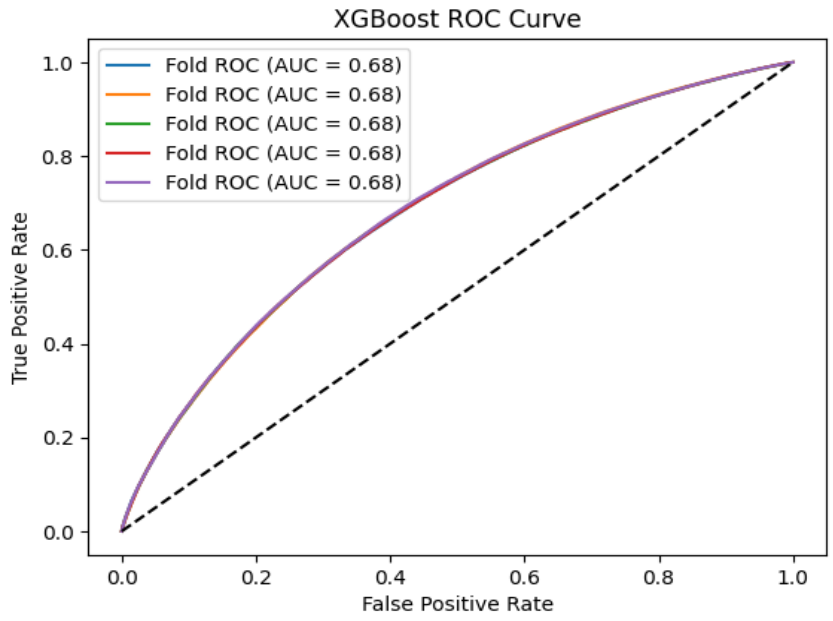

Fig 19: XGBoost ROC Curve

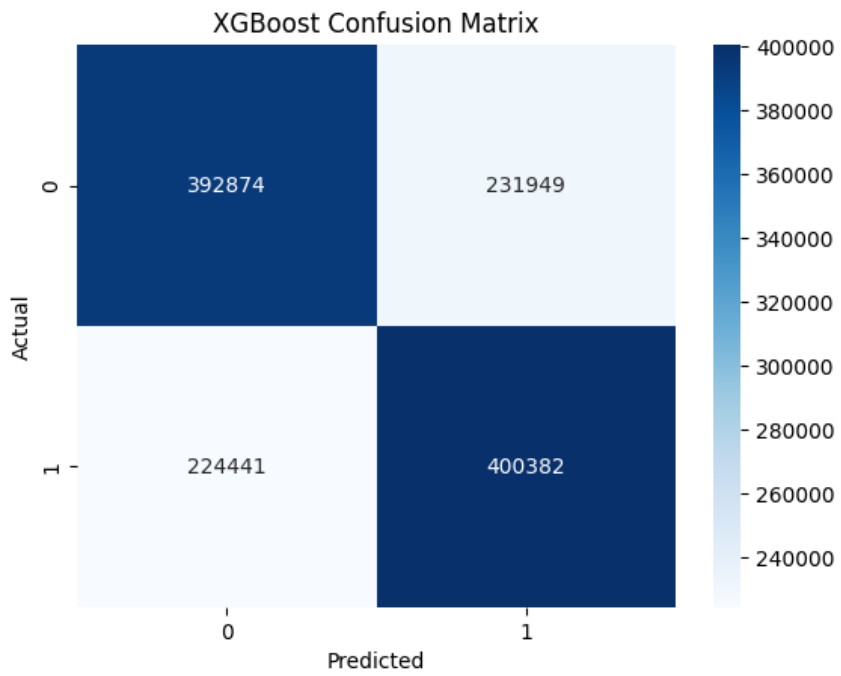

Fig 20: XGBoost Confusion Matrix

5.5 Logistic regression

Logistic regression is a widely used linear model for binary classification problems. Particularly in cases with primarily linear relationships between variables, it is preferred for its simplicity, ease of implementation and interpretability. Logistic regression was performed in this study with an accuracy of 62%, correctly classifying 62% of the cases. The model's ROC AUC score of 0.67 shows moderate ability to discriminate between positive and negative classifications. While the recall score of 65% indicates that it effectively spotted 65% of the real positive cases, the model accurately identified 62% of its positive predictions with a precision of 62%. The F1 score of 0.63 reflects a fair balance between precision and recall, but overall, the model's performance lags more complex ensemble models like Random Forest and XGBoost. The confusion matrix shows that logistic regression generated 253,029 false positive and 219, 850 false negatives, therefore suggesting a greater rate of misclassification compared to the ensemble models. Logistic regression is less appropriate for this dataset even if it is straightforward to understand and apply

as its performance in terms of accuracy and ROC AUC suffers when compared to more advanced models.

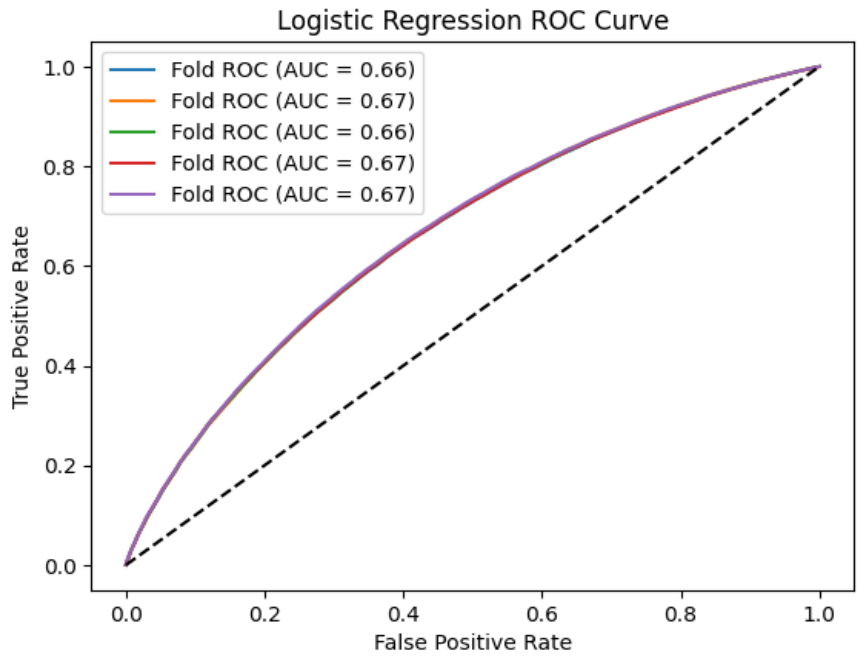

Fig 21: Logistic regression ROC Curve

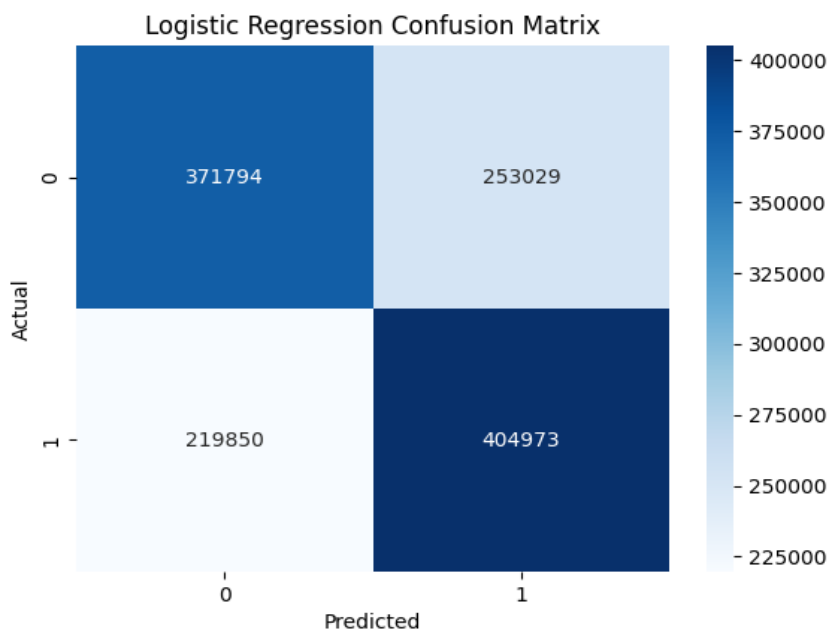

Fig 22: Logistic regression Confusion Matrix

5.6 Deep Neural Network (DNN)

Deep neural networks (DNN) are complex models designed to grasp complex non-linear connections in data. They are particularly useful when managing large datasets or when feature interactions are critical to the task at hand. However, these models demand substantial computational resources and require large amounts of data to train effectively. In this analysis, the DNN model underperformed, achieving an accuracy of only 52%, which states that nearly half of the predictions were incorrect. Its ROC AUC score of 0.53 suggests that the model struggled to differentiate positive from negative classes, performing only marginally better than random guessing. The precision of 33%, with just a third of the positive forecasts accurate, demonstrates low precision. The recall score of 51% shows that the model was able to find somewhat over half of the real positive cases. The F1 score of 0.39 reflects an imbalance between recall and precision, so stressing poor general performance. The confusion matrix reveals even more notable misclassification, with 291,831 false positives and 303,758 false negatives, the highest among all the models tested. These findings imply that the DNN model failed with this dataset, mostly likely because of its complexity and possible overfitting problems, which would have resulted in a far weaker performance than simpler models like Random Forest and XGBoost.

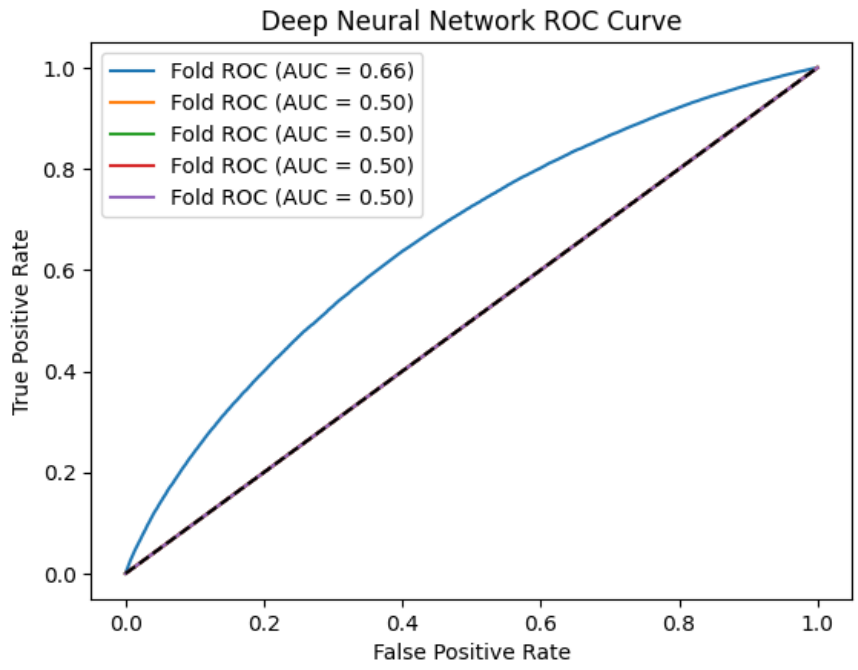

Fig 23: Deep Neural Network ROC Curve

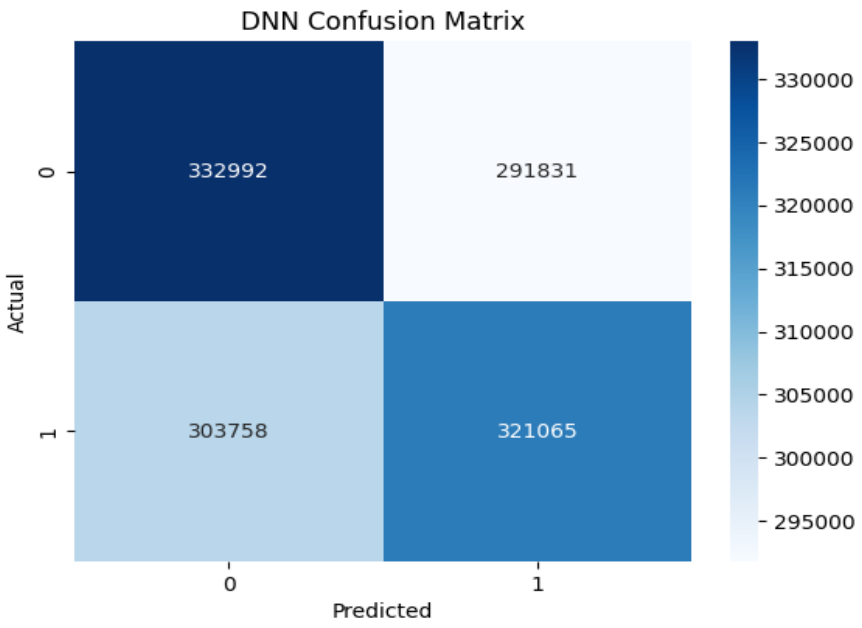

Fig 24: DNN Confusion Matrix

Random Forest showed among the models assessed the best performance across all major criteria with the maximum accuracy of 73% and a robust ROC AUC score of 0.81. This suggests that it was most likely able to differentiate between positive and negative classifications. Random Forest also showed a well-balanced trade-off between accuracy, recall, and F1- score., therefore verifying its position as the most successful model for this dataset. Both LightGBM and XGBoost displayed moderate performance with comparable accuracy scores around 63% and ROC AUC values of 0.68. Particularly in terms of general accuracy and discriminative capacity, these gradient boosting models proved inadequate when compared to Random Forest even if they showed good predictive potential. Logistic regression, though simpler and more interpretable, produced reasonable results with an accuracy of 62% and a ROC AUC of 0.67, but it could not match the performance of the ensemble models. Finally, the Deep Neural Network (DNN) battled the most, yielding the lowest accuracy at 52% and an ROC AUC of 0.53, indicating it was barely better than random guessing. DNN's poor precision, recall, and F1-score highlight even more its unsuitability for this dataset. Overall, Random Forest emerged as the best model, outperforming the other techniques in terms of accuracy, balance and predictive power.

Random forest proved the best performance for this lung cancer dataset based on the examination of five machine learning models, so it is the most appropriate model for next research. Finding out which clinical and non-clinical factors most affect lung cancer death comes next. Understanding the causes of lung cancer outcomes and maybe guiding treatment and preventive measures depends on this study. Random forest, with its ability to rank feature importance, is used here to evaluate the impact of both clinical and non-clinical factors on lung cancer mortality.

	Random Forest	LightGBM	XGBoost	Logistic Regression	DNN
Accuracy	0.73	0.63	0.63	0.62	0.52
ROC AUC	0.81	0.68	0.68	0.67	0.53
Precision	0.73	0.63	0.63	0.62	0.33
Recall	0.74	0.63	0.64	0.65	0.51
F1 Score	0.73	0.63	0.64	0.63	0.39

Table 2: Overall performance of five models

5.7 Influence of Clinical and non-clinical factors in lung cancer mortality

In this study, lung cancer mortality was evaluated using both clinical and non-clinical factors. Clinical factors such as age, sex, comorbidities, cancer stage, preoperative radiation, extent of resection, positive surgical margins, and tumor size were linked to 30-day mortality following resection. Additionally, non-clinical factors including residing in lower income neighborhoods with fewer high school graduates and receiving cancer treatment at nonacademic medical centers were independently associated with higher 30-day postoperative mortality [31]. According to the above research, Clinical variables directly pertaining to the patient's medical state were age, gender, cancer stage, BMI, cholesterol levels, hypertension, asthma, cirrhosis and other cancer history. These elements are supposed to be rather important for mortality. Though not explicitly

medical, non-clinical elements such family history, treatment type and smoking status can still influence health outcomes. When Random Forest model performed highly when applied using only clinical factors, achieving an accuracy of 0.73 and ROC AUC of 0.80, reflecting a good ability to differentiate between survivors and non-survivors. Based on clinical characteristics, precision and recall values of 0.72 and 0.74 respectively show even more how well the model forecasts lung cancer on death. On the other hand, the performance declined dramatically when the model was used with only non-clinical factors, with accuracy of 0.51 and a ROC AUC of 0.52, thereby showing inadequate discriminating between outcomes. The precision and recall were both 0.51, reflecting lesser predictive power. These findings highlight the need of clinical factors in precisely estimating lung cancer mortality; non- clinical factors, although important, lack the predictive power required on their alone. The two model's extreme performance differences draw attention to the crucial part medical data performs in lung cancer patient outcome prediction.

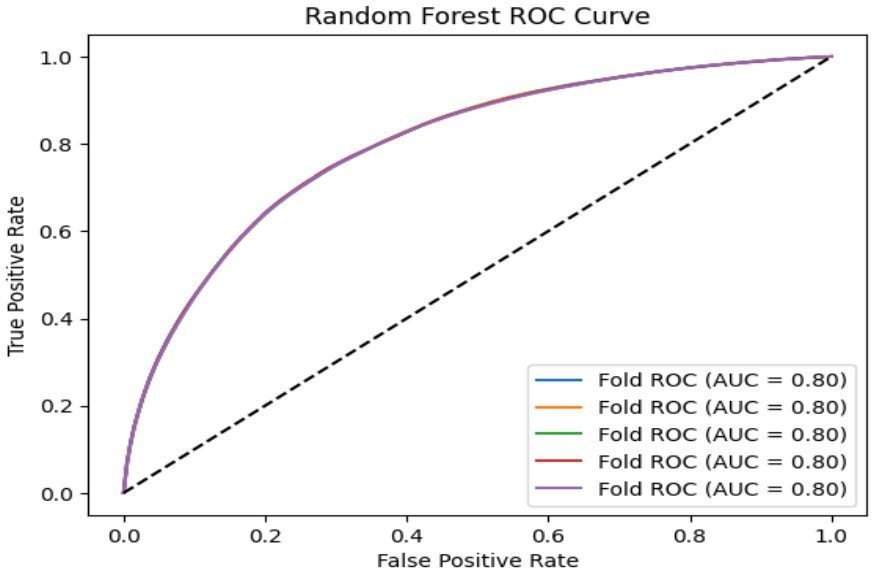

Fig 25: Random Forest ROC Curve for Clinical factors

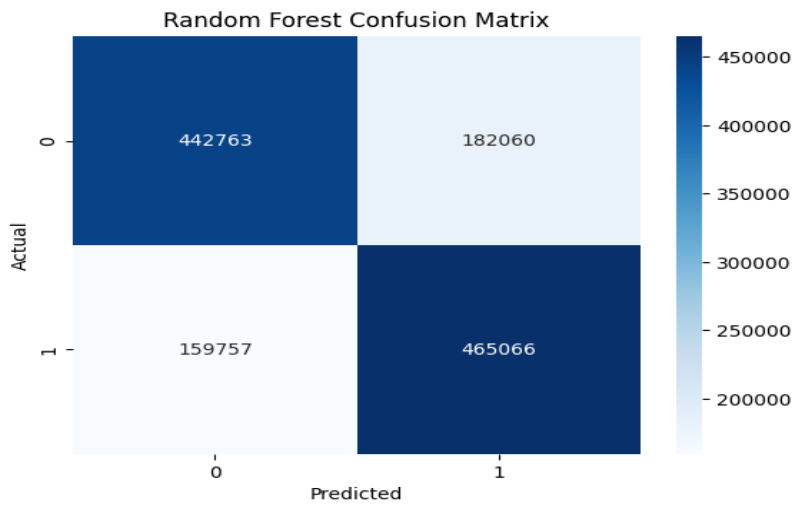

Fig 26: Random Forest Confusion Matrix for Clinical factors

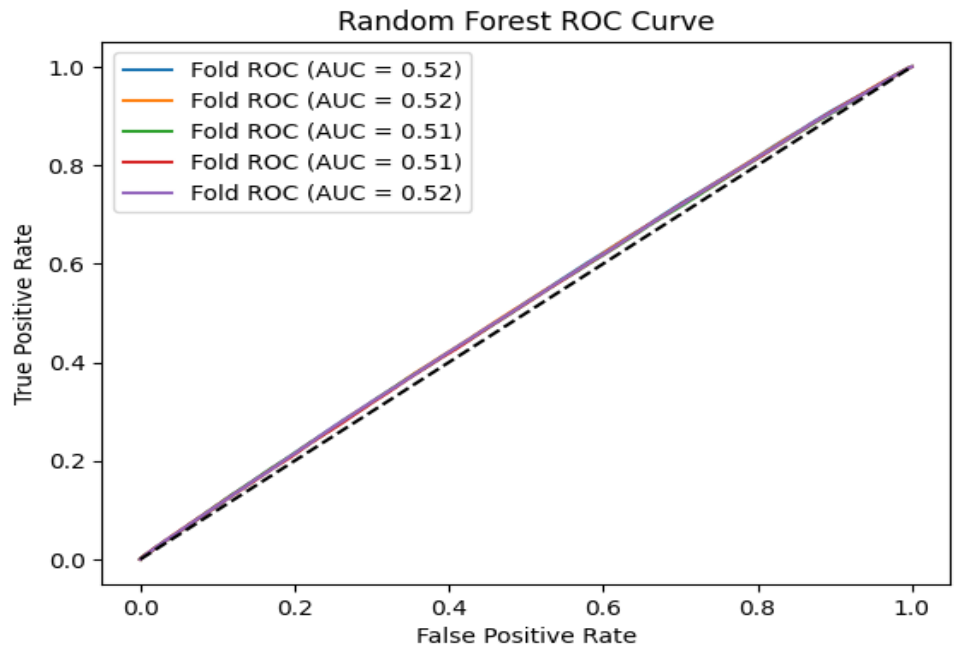

Fig 27: Random Forest ROC Curve for Non-Clinical Factors

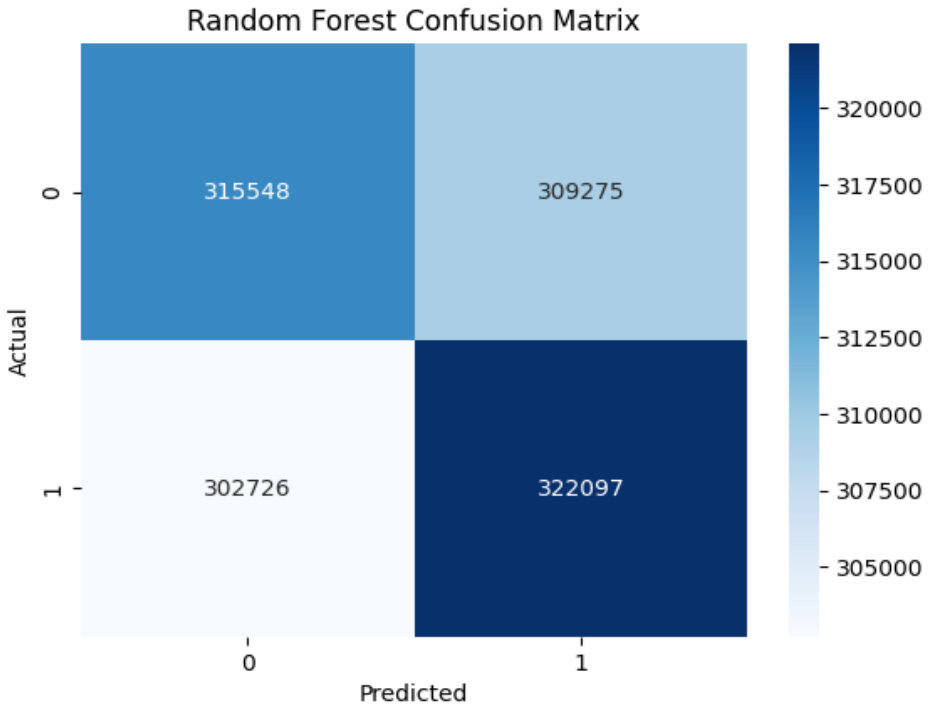

Fig 28: Random Forest Confusion Matrix for Non-Clinical Factors

CHAPTER SIX

Conclusion and Recommendation for Future work

6.1 Conclusion

This study sought to use a large dataset of over 3.25 million records from 27 European countries to find the most accurate predictive model for lung cancer mortality. The dataset was comprised of 17 explanatory variables, including both clinical and non-clinical factors, utilized to create models able to forecast patient survival. Several machine learning models were evaluated, such as Random Forest, LightGBM, XGBoost, Logistic regression, and Deep Neural Networks (DNN). Among these, Random Forest shows the greatest performance with an accuracy of 73%, a ROC AUC of 0.81, and a well-balanced precision and recall. It was the best model for this dataset since it had the best capacity to separate surviving from non-surviving patients. Models like LightGBM, XGBoost and Logistic regression performed moderately well but were outperformed by Random Forest, while the DNN model showed the lowest results, most likely due to overfitting and data complexity.

The study showed that lung cancer survival was determined mainly by clinical variables including cancer stage, BMI, Cholesterol levels, hypertension, asthma, cirrhosis and treatment type. These variables contributed significantly to the prediction accuracy of the model, while non-clinical factors like age, gender, family history, and smoking status, however instructive, had poor predictive ability when employed alone. This emphasizes the importance of medical data in improving the precision of lung cancer mortality predictions.

6.2 Recommendation for future work

Incorporation of additional variables: future research could add more clinical variables such genetic makers, family medical history or comorbidities. These factors might offer more thorough understanding of lung cancer development and survival, hence improving the models' predictive capacity.

Advanced feature engineering: these approaches include interaction terms or polynomial features that could be used to capture more complicated connections between variables. Furthermore, applying more advanced features selection techniques could help to find the most significant predictors, hence perhaps lowering model complexity and raising accuracy.

Exploring different sampling techniques: While SMOTE handled class imbalance, investigating various resampling strategies including Adaptive Synthetic Sampling (ADASYN) or NearMiss could help to further improve model performance. Combining under sampling and over sampling approaches in hybrid ways could yield more consistent and strong findings.

Testing other machine learning models: Future research could focus on evaluating more sophisticated machine learning models, including CatBoost or ensemble techniques like staking, which aggregates several models to attain superior predictive capability. These methods might enable the generalizing capacity of the model to fresh data and help to increase accuracy.

Temporal analysis and survival models: Applying survival approaches such Cox Proportional Hazards models or time- series models could produce more dynamic forecasts considering the longitudinal character of the data (including dates of diagnosis and treatment). Particularly helpful in medical prognosis models may explain variations over time.

Model interpretability and Explainability: Though Random Forest performed the best; its ensemble character makes it less interpretable. Future research could investigate explainability methods as LIME (Local Interpretable Model-agnostic Explanations) or SHAP (Shapley Additive exPlanations) to better grasp the influence of individual variables on the prediction, so important in clinical decision-making.

Real-world validation: Validation of these models from outside geographic areas or healthcare systems is crucial in the real-world sense. This would improve the model for more varied patient populations and assist to evaluate the generalizability of the results.

Integration with health care systems: the generated prediction model in this work could perhaps be included into systems of healthcare decision-support. Future research could include creating an automated tool for doctors to evaluate patient traits-based lung cancer risk and provide individualized treatment approaches.

Extensive study in these fields will help the predictive models for lung cancer survival to become more accurate, interpretable, and relevant in real- world clinical environments, thereby improving patient outcomes and more sensible healthcare policy.

Google Colab link for python coding

https://colab.research.google.com/drive/1qXnu0V8zJ-k-SsYeHhWqE70Fd3ZuFU1p?usp=sharing

https://colab.research.google.com/drive/1H69chlGoDh12Fc0tkX5WynimS2HM5NSj?usp=sharing

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Appendix

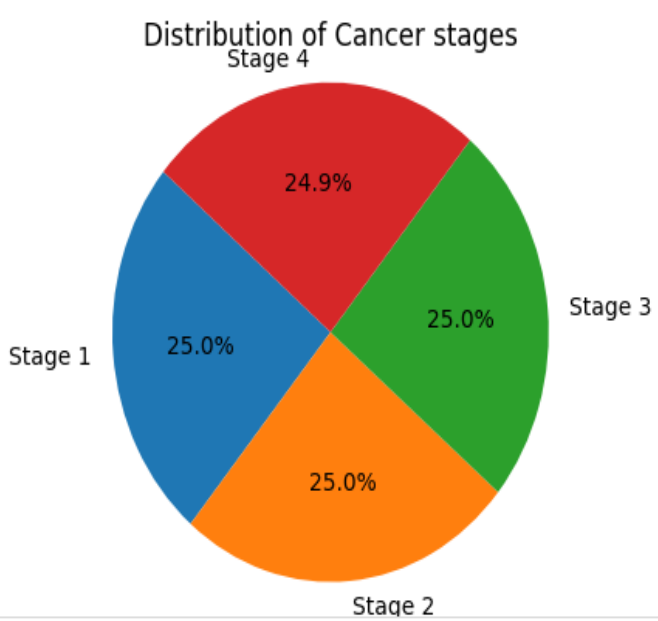
Exploratory Analysis

```python
import pandas as pd
data=pd.read_csv("/content/drive/MyDrive/lung cancer.csv")
print(data)
```

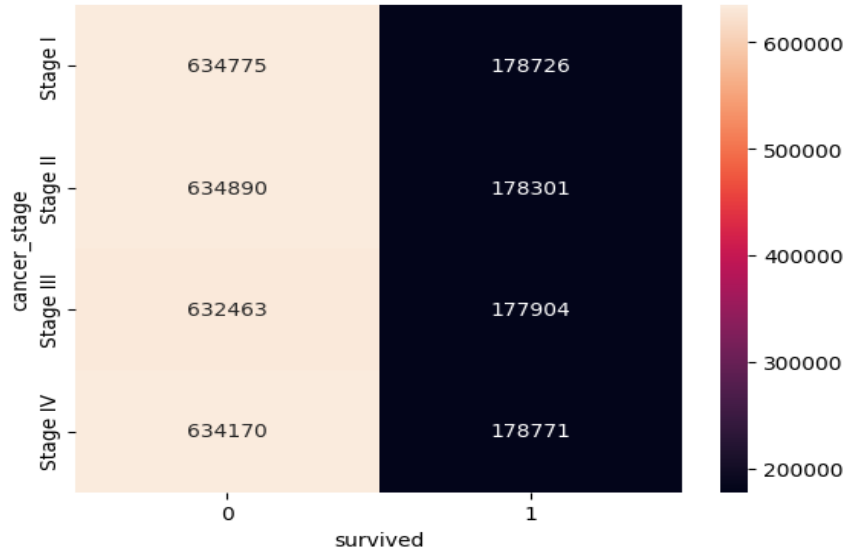
```
              id   age  gender         country diagnosis_date cancer_stage  \
0              1  64.0  Female         Germany     2016-04-07    Stage III
1              2  50.0    Male  Czech Republic     2023-04-22    Stage III
2              3  65.0    Male         Romania     2023-04-07     Stage IV
3              4  51.0  Female          Latvia     2016-02-07    Stage III
4              5  37.0    Male          Greece     2023-12-01      Stage I
...          ...   ...     ...             ...            ...          ...
3249995  3249996  45.0    Male          Poland     2020-03-11     Stage IV
3249996  3249997  51.0    Male           Italy     2016-11-16     Stage IV
3249997  3249998  45.0    Male         Finland     2021-07-18    Stage III
3249998  3249999  45.0    Male         Hungary     2020-07-25      Stage I
3249999  3250000  59.0    Male         Croatia     2018-05-12    Stage III

        beginning_of_treatment_date family_history  smoking_status   bmi  \
0                        2016-04-21             No    Never Smoked  31.1
1                        2023-05-02            Yes  Passive Smoker  25.9
2                        2023-04-12             No    Never Smoked  18.9
3                        2016-02-13            Yes   Former Smoker  34.6
4                        2023-12-03            Yes    Never Smoked  40.2
...                             ...            ...             ...   ...
3249995                  2020-03-18             No  Current Smoker  31.0
3249996                  2016-11-22            Yes  Current Smoker  41.2
3249997                  2021-07-30            Yes  Current Smoker  18.2
3249998                  2020-07-30             No  Passive Smoker  40.2
3249999                  2018-05-20             No   Former Smoker  25.4

         cholesterol_level  hypertension  asthma  cirrhosis  other_cancer  \
0                      257             1       1          0             0
1                      208             1       0          0             0
2                      193             0       0          0             0
3                      249             1       1          1             0
4                      262             0       0          0             0
...                    ...           ...     ...        ...           ...
3249995                273             1       1          0             0
3249996                279             1       1          0             0
3249997                187             1       1          0             0
3249998                271             1       0          0             0
```

```
         cholesterol_level  hypertension  asthma  cirrhosis  other_cancer  \
0                      257             1       1          0             0
1                      208             1       0          0             0
2                      193             0       0          0             0
3                      249             1       1          1             0
4                      262             0       0          0             0
...                    ...           ...     ...        ...           ...
3249995                273             1       1          0             0
3249996                279             1       1          0             0
3249997                187             1       1          0             0
3249998                271             1       0          0             0
3249999                218             0       1          0             0

        treatment_type end_treatment_date  survived
0             Combined         2017-11-15         0
1            Radiation         2024-04-25         0
2              Surgery         2025-03-11         0
3              Surgery         2017-04-14         1
4          Chemotherapy         2024-09-20         0
...                 ...                ...       ...
3249995    Chemotherapy         2020-11-26         1
3249996          Surgery         2018-04-10         0
3249997         Combined         2022-03-16         0
3249998          Surgery         2022-02-28         0
3249999          Surgery         2019-06-01         0

[3250000 rows x 18 columns]
```

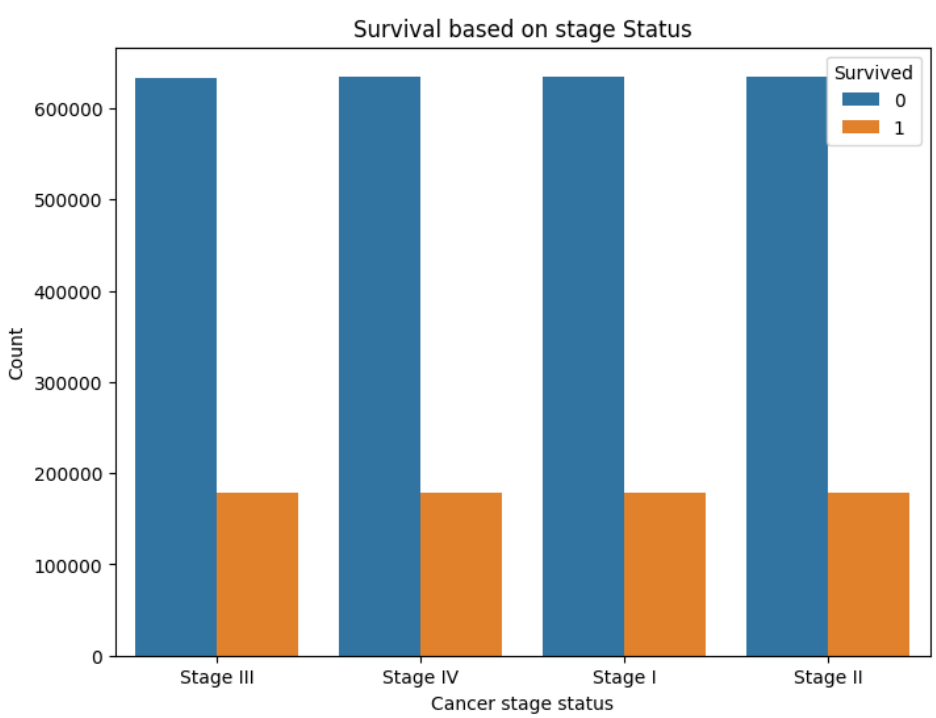
```python
import pandas as pd
import matplotlib.pyplot as plt

# Count the occurrences of each cancer stage
cancer_stage_counts = data['cancer_stage'].value_counts()

# Define labels for the different cancer stage
labels = ['Stage 1', 'Stage 2', 'Stage 3', 'Stage 4']

# Plotting the pie chart
plt.figure(figsize=(4, 4))
plt.pie(cancer_stage_counts, labels=labels, autopct='%1.1f%%', startangle=140)

# Adding title
plt.title('Distribution of Cancer stages')

# Displaying the plot
plt.axis('equal')
plt.show()
```



```
[ ] import seaborn as sns
    cross = pd.crosstab(data["cancer_stage"], data["survived"])
    sns.heatmap(cross, annot=True, fmt='d')
    plt.show()
```

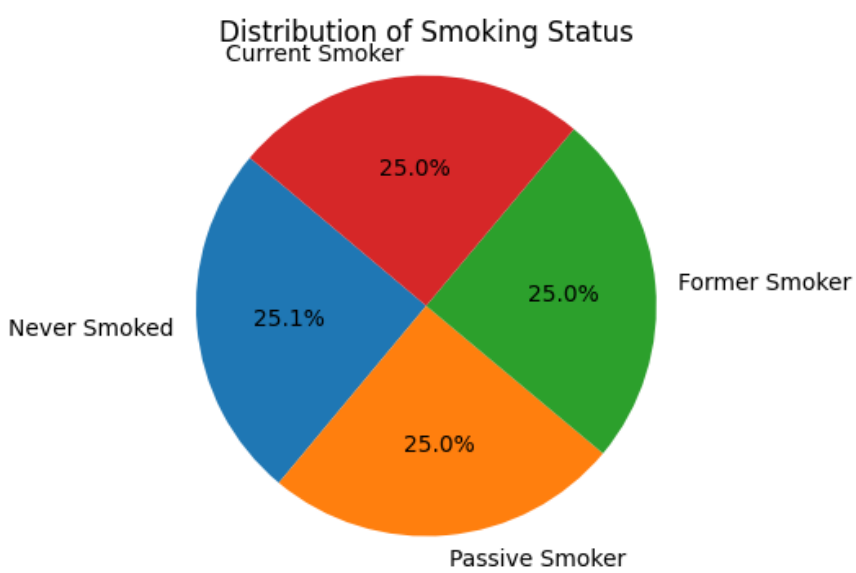



```
plt.figure(figsize=(8, 6))
sns.countplot(data=data, x='cancer_stage', hue='survived')
plt.title('Survival based on stage Status')
plt.xlabel('Cancer stage status')
plt.ylabel('Count')
plt.legend(title='Survived', loc='upper right')
plt.show()
```

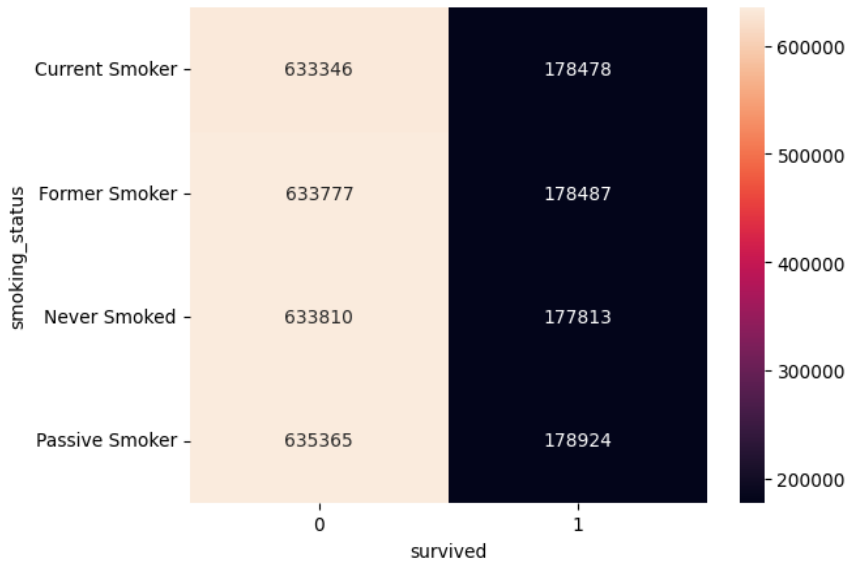


```
[ ] SMOKING STATUS
```

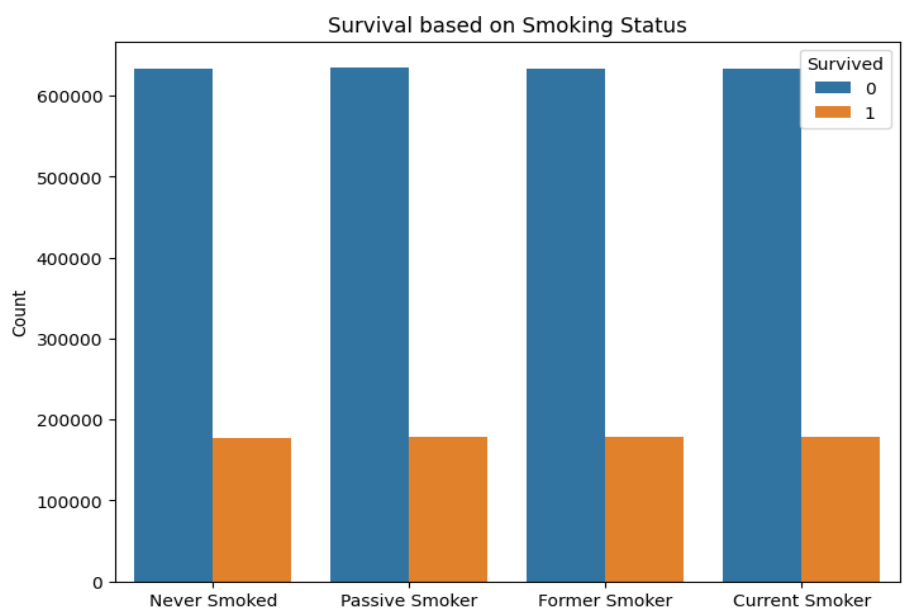
```python
# Count the occurrences of each smoking status
smoking_status_counts = data['smoking_status'].value_counts()

# Define labels for the different smoking status
labels = ['Never Smoked', 'Passive Smoker', 'Former Smoker', 'Current Smoker']

# Plotting the pie chart
plt.figure(figsize=(4, 4))
plt.pie(smoking_status_counts, labels=labels, autopct='%1.1f%%', startangle=140)

# Adding title
plt.title('Distribution of Smoking Status')

# Displaying the plot
plt.axis('equal')
plt.show()
```



```python
import seaborn as sns
cross = pd.crosstab(data["smoking_status"], data["survived"])
sns.heatmap(cross, annot=True, fmt='d')
plt.show()
```

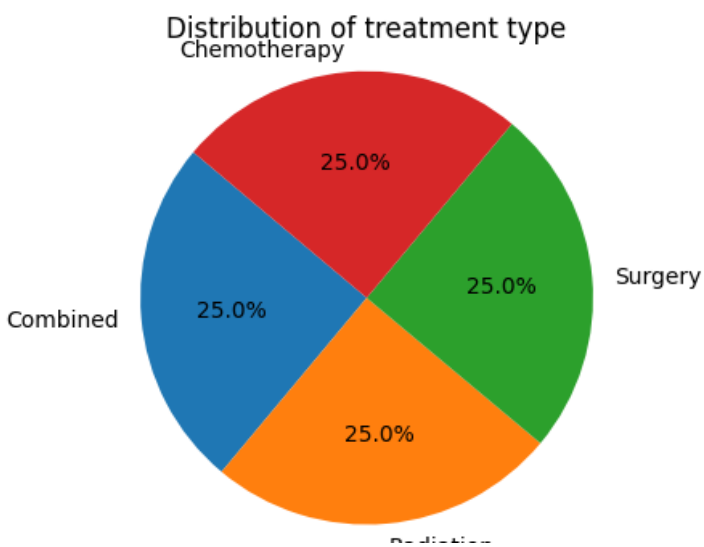



```python
plt.figure(figsize=(8, 6))
sns.countplot(data=data, x='smoking_status', hue='survived')
plt.title('Survival based on Smoking Status')
plt.xlabel('Smoking Status')
plt.ylabel('Count')
plt.legend(title='Survived', loc='upper right')
plt.show()
```

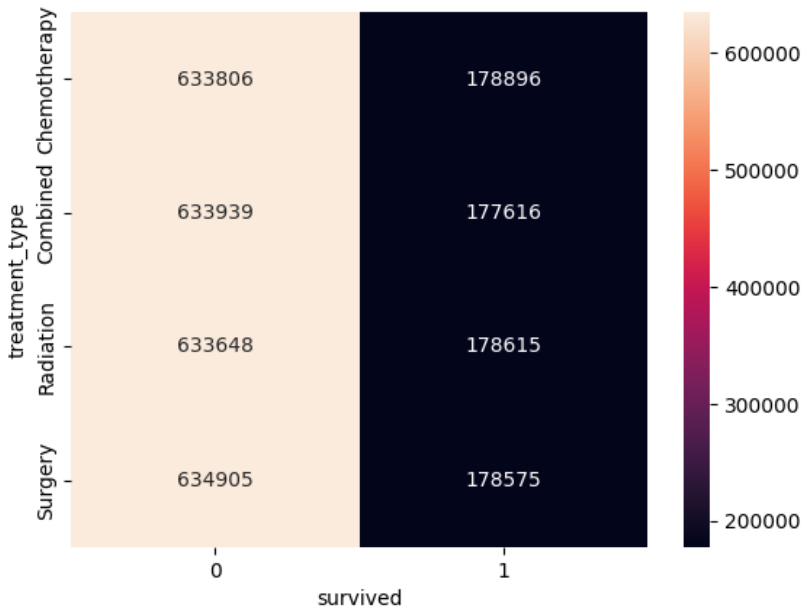


```
[ ]  TREATMENT TYPE
```

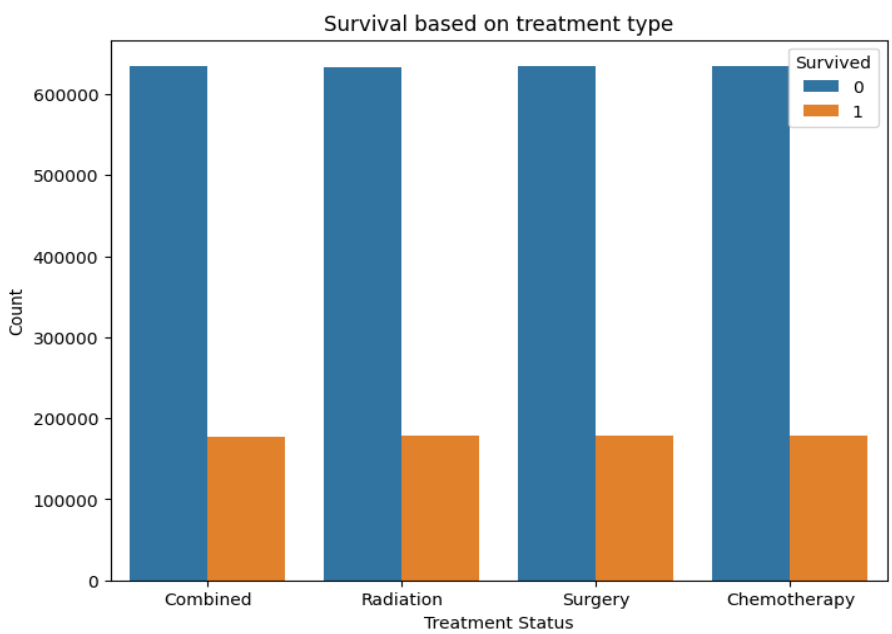
```python
# Count the occurrences of each treatment type
treatment_type_counts = data['treatment_type'].value_counts()

# Define labels for the different treatment type
labels = ['Combined', 'Radiation', 'Surgery', 'Chemotherapy']

# Plotting the pie chart
plt.figure(figsize=(4, 4))
plt.pie(treatment_type_counts, labels=labels, autopct='%1.1f%%', startangle=140)

# Adding title
plt.title('Distribution of treatment type')

# Displaying the plot
plt.axis('equal')
plt.show()
```



```
cross = pd.crosstab(data["treatment_type"], data["survived"])
sns.heatmap(cross, annot=True, fmt='d')
plt.show()
```

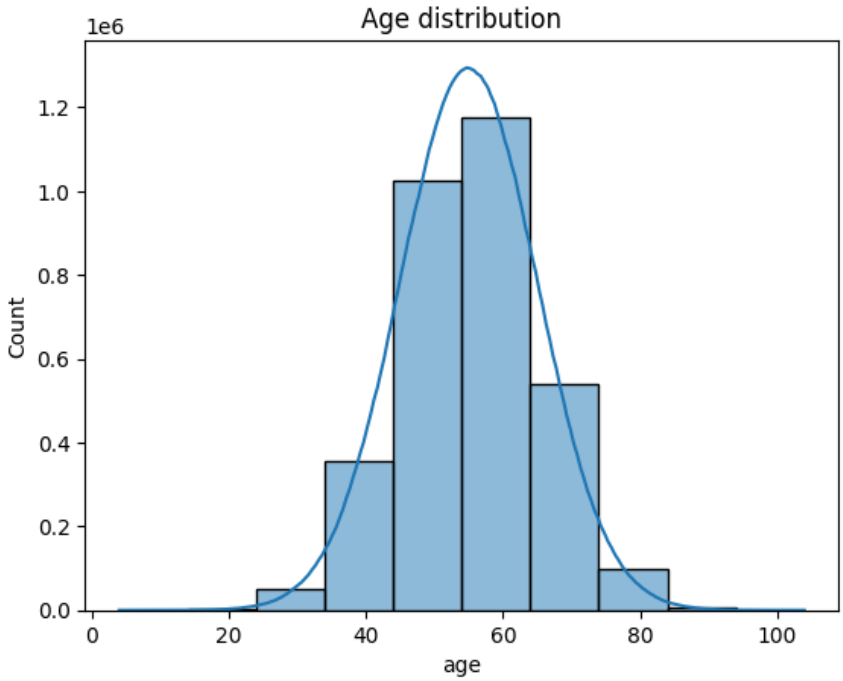



```
plt.figure(figsize=(8, 6))
sns.countplot(data=data, x='treatment_type', hue='survived')
plt.title('Survival based on treatment type')
plt.xlabel('Treatment Status')
plt.ylabel('Count')
plt.legend(title='Survived', loc='upper right')
plt.show()
```



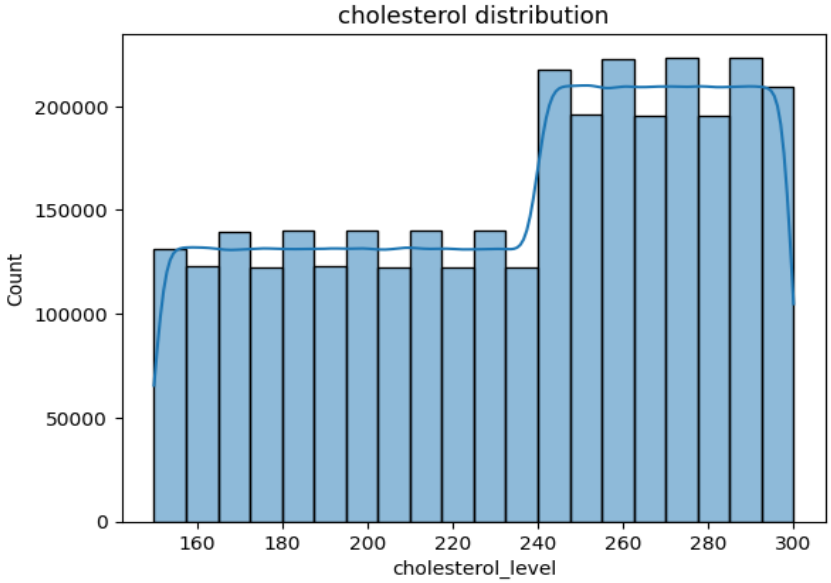
```
[ ] sns.histplot(data['age'] , bins = 10 , kde = True)
    plt.title('Age distribution')
    plt.plot()
```

```
[]
```

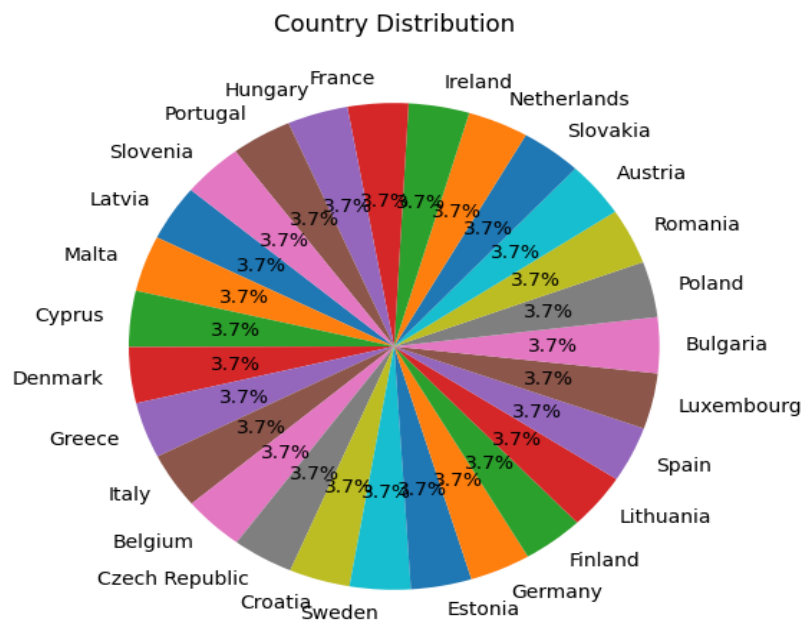



```
sns.histplot(data['cholesterol_level'] , bins = 20 , kde = True)
plt.title('cholesterol distribution')
plt.plot()
```

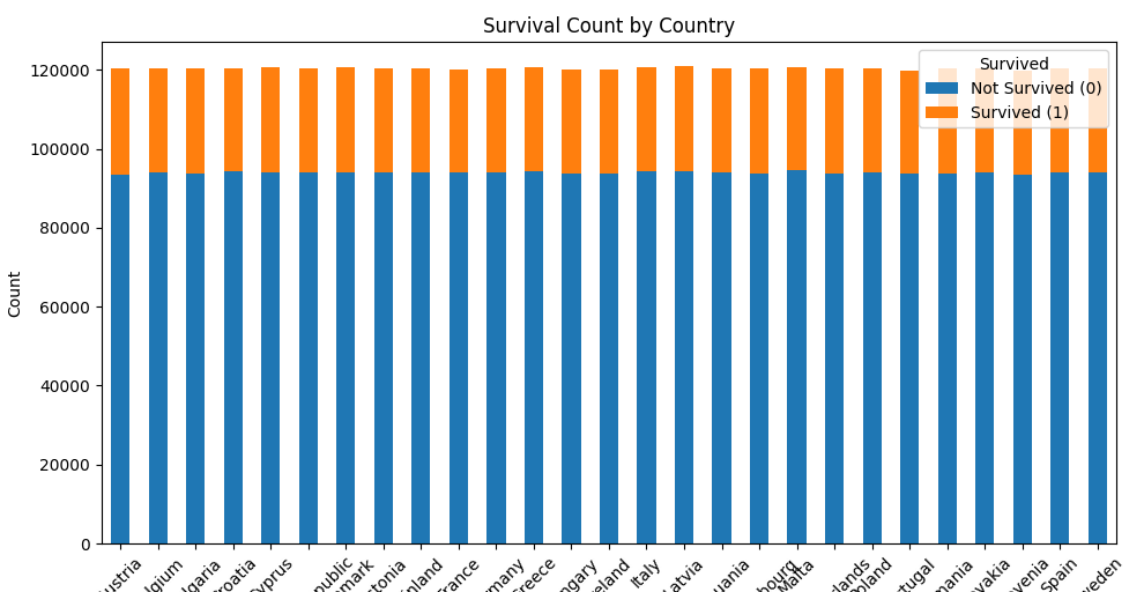
```
[]
```



```
ct = data['country'].value_counts()
plt.figure(figsize=(6, 6))
plt.pie(data['country'].value_counts(), labels=ct.index, autopct='%1.1f%%', startangle=140)
plt.title('Country Distribution')
plt.show()
```



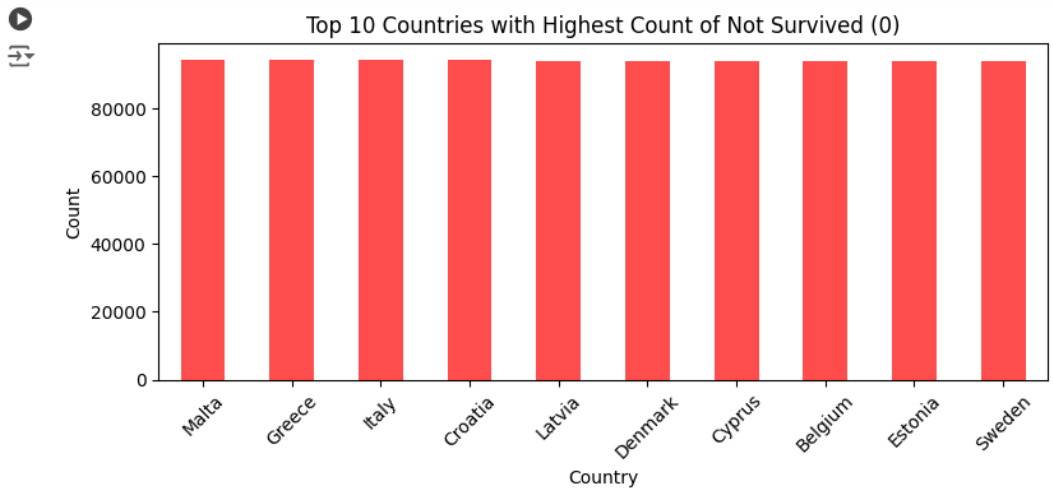

```
grouped_data = data.groupby(['country', 'survived']).size().unstack()

# Plotting the bar chart
grouped_data.plot(kind='bar', stacked=True, figsize=(10, 6))
plt.xlabel('Country')
plt.ylabel('Count')
plt.title('Survival Count by Country')
plt.legend(title='Survived', labels=['Not Survived (0)', 'Survived (1)'])
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

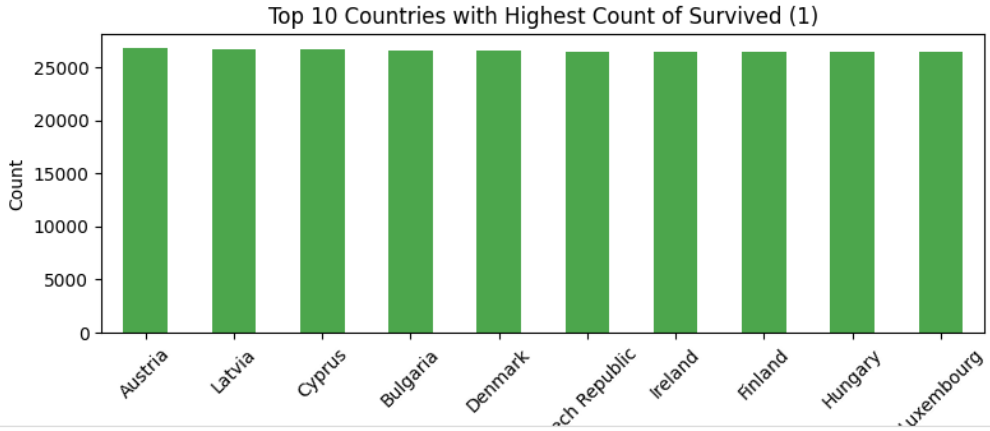


```
[ ] # Printing the grouped data (value count by country and survival status)
    print(grouped_data)
```

```
survived            0      1
country
Austria         93474  26778
Belgium         94052  26435
Bulgaria        93663  26633
Croatia         94214  26259
Cyprus          94089  26703
Czech Republic  93953  26525
Denmark         94133  26575
Estonia         94037  26351
Finland         93880  26478
France          93887  26293
Germany         93989  26374
Greece          94294  26365
Hungary         93644  26476
Ireland         93721  26484
Italy           94223  26382
Latvia          94205  26742
Lithuania       94020  26334
Luxembourg      93844  26456
Malta           94444  26349
Netherlands     93791  26427
Poland          93951  26343
Portugal        93674  26133
Romania         93825  26452
Slovakia        93906  26327
Slovenia        93408  26229
Spain           93950  26395
Sweden          94027  26404
```

```
# Grouping data by country and survival status
grouped_data = data.groupby(['country', 'survived']).size().unstack(fill_value=0)

# Sorting the data to get the top 5 countries with the highest count of 0 (not survived) and 1 (survived)
top_10_not_survived = grouped_data[0].sort_values(ascending=False).head(10)
top_10_survived = grouped_data[1].sort_values(ascending=False).head(10)

# Plotting the bar chart for top 5 countries with highest count of 0 (not survived)
plt.figure(figsize=(8, 4))
top_10_not_survived.plot(kind='bar', color='red', alpha=0.7)
plt.xlabel('Country')
plt.ylabel('Count')
plt.title('Top 10 Countries with Highest Count of Not Survived (0)')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Plotting the bar chart for top 5 countries with highest count of 1 (survived)
plt.figure(figsize=(8, 4))
top_10_survived.plot(kind='bar', color='green', alpha=0.7)
plt.xlabel('Country')
plt.ylabel('Count')
plt.title('Top 10 Countries with Highest Count of Survived (1)')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```





Connected to Python 3 Google Compute Engine backend

Predictive analysis

```python
import pandas as pd
data=pd.read_csv("/content/drive/MyDrive/Lu.cancer.csv")
print(data)
```

```
             id  age  gender         country diagnosis_date cancer_stage  \
0             1   64  Female         Germany     07/04/2016    Stage III
1             2   50    Male  Czech Republic     22/04/2023    Stage III
2             3   65    Male         Romania     07/04/2023     Stage IV
3             4   51  Female          Latvia     07/02/2016    Stage III
4             5   37    Male          Greece     01/12/2023      Stage I
...         ...  ...     ...             ...            ...          ...
999995   999996   56    Male          Cyprus     08/11/2018     Stage II
999996   999997   55    Male          France     21/07/2014    Stage III
999997   999998   49    Male         Romania     21/02/2021     Stage IV
999998   999999   61    Male         Austria     23/10/2014     Stage II
999999  1000000   53  Female        Slovenia     04/02/2021      Stage I

       beginning_of_treatment_date family_history  smoking_status   bmi  \
0                       21/04/2016             No    Never Smoked  31.1
1                       02/05/2023            Yes  Passive Smoker  25.9
2                       12/04/2023             No    Never Smoked  18.9
3                       13/02/2016            Yes   Former Smoker  34.6
4                       03/12/2023            Yes    Never Smoked  40.2
...                            ...            ...             ...   ...
999995                  14/11/2018             No    Never Smoked  43.2
999996                  26/07/2014             No  Passive Smoker  34.2
999997                  23/02/2021             No  Current Smoker  37.0
999998                  10/11/2014             No  Current Smoker  16.1
999999                  18/02/2021             No    Never Smoked  39.9

        cholesterol_level  hypertension  asthma  cirrhosis  other_cancer  \
0                     257             1       1          0             0
1                     208             1       0          0             0
2                     193             0       0          0             0
3                     249             1       1          1             0
4                     262             0       0          0             0
...                   ...           ...     ...        ...           ...
999995                244             1       0          0             0
999996                260             0       0          0             0
999997                264             0       0          0             0
999998                177             0       0          0             0
```

```
        cholesterol_level  hypertension  asthma  cirrhosis  other_cancer  \
0                     257             1       1          0             0
1                     208             1       0          0             0
2                     193             0       0          0             0
3                     249             1       1          1             0
4                     262             0       0          0             0
...                   ...           ...     ...        ...           ...
999995                244             1       0          0             0
999996                260             0       0          0             0
999997                264             0       0          0             0
999998                177             0       0          0             0
999999                261             1       1          0             0

       treatment_type end_treatment_date  survived
0            Combined         15/11/2017         0
1           Radiation         25/04/2024         0
2             Surgery         11/03/2025         0
3             Surgery         14/04/2017         1
4        Chemotherapy         20/09/2024         0
...               ...                ...       ...
999995       Combined         13/11/2019         0
999996      Radiation         14/02/2015         0
999997   Chemotherapy         27/03/2022         0
999998       Combined         08/12/2015         1
999999      Radiation         05/01/2023         0

[1000000 rows x 18 columns]
```

```
data['gender'] = data['gender'].map({'Male': 1, 'Female': 0})
data['cancer_stage'] = data['cancer_stage'].map({'Stage I': 1, 'Stage II': 2, 'Stage III':3, 'Stage IV':4})
data['treatment_type'] = data['treatment_type'].map({'Chemotherapy': 1, 'Combined': 2, 'Radiation':3, 'Surgery':4})
data['smoking_status'] = data['smoking_status'].map({'Current Smoker': 1, 'Former Smoker': 2, 'Never Smoked':3, 'Passive Smoker':4})
data['family_history'] = data['family_history'].map({'Yes': 1, 'No': 0})
```

```
# Import necessary libraries
from imblearn.over_sampling import SMOTE
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# Define your features (X) and target variable (y)
X = data[['age', 'gender', 'cancer_stage', 'family_history', 'smoking_status', 'bmi', 'cholesterol_level', 'hypertension', 'asthma', 'cirrhosis', 'other_cancer', 'treatment_type']]
y = data['survived']

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Apply SMOTE to the training data
smote = SMOTE(random_state=42)
X_train_res, y_train_res = smote.fit_resample(X_train, y_train)


# Check the distribution of the resampled training data
print("Training set after SMOTE:")
print(y_train_res.value_counts())

# Check the distribution of the original testing data
print("\nTesting set distribution:")
print(y_test.value_counts())
```

```
        cholesterol_level  hypertension  asthma  cirrhosis  other_cancer  \
0                     257             1       1          0             0
1                     208             1       0          0             0
2                     193             0       0          0             0
3                     249             1       1          1             0
4                     262             0       0          0             0
...                   ...           ...     ...        ...           ...
999995                244             1       0          0             0
999996                260             0       0          0             0
999997                264             0       0          0             0
999998                177             0       0          0             0
999999                261             1       1          0             0

       treatment_type end_treatment_date  survived
0            Combined         15/11/2017         0
1           Radiation         25/04/2024         0
2             Surgery         11/03/2025         0
3             Surgery         14/04/2017         1
```

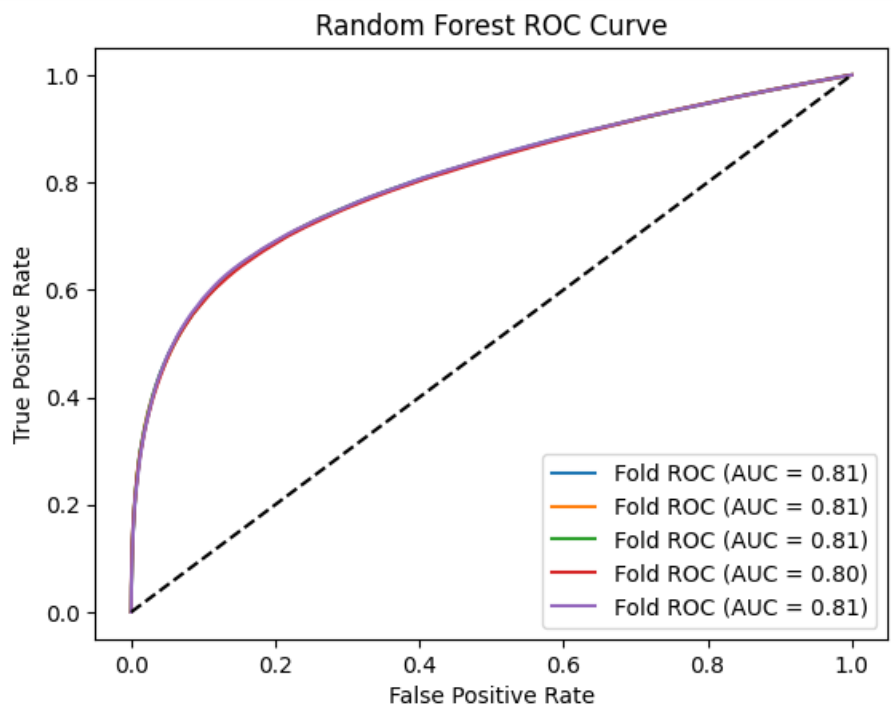
```
[ ] Training set after SMOTE:
    survived
    1    624823
    0    624823
    Name: count, dtype: int64

    Testing set distribution:
    survived
    0    155718
    1     44282
    Name: count, dtype: int64
```

```python
from sklearn.ensemble import RandomForestClassifier
from lightgbm import LGBMClassifier
from xgboost import XGBClassifier
from sklearn.metrics import accuracy_score, roc_auc_score, roc_curve
from sklearn.model_selection import StratifiedKFold
import matplotlib.pyplot as plt
import numpy as np
from sklearn.metrics import precision_score, recall_score, f1_score, confusion_matrix, classification_report
import seaborn as sns
from sklearn.linear_model import LogisticRegression
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout
```

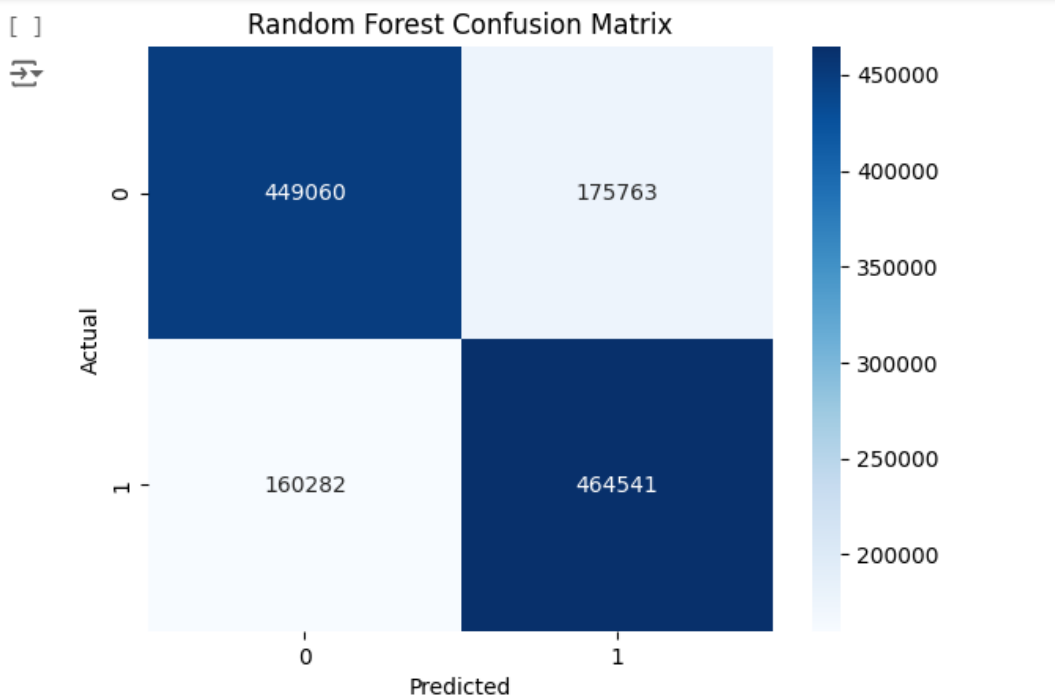
```
/usr/local/lib/python3.10/dist-packages/dask/dataframe/__init__.py:42: FutureWarning:
Dask dataframe query planning is disabled because dask-expr is not installed.

You can install it with `pip install dask[dataframe]` or `conda install dask`.
This will raise in a future version.

  warnings.warn(msg, FutureWarning)
```

```python
# Initialize StratifiedKFold
kf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

# Initialize lists to store results
rf_accuracy = []
rf_roc_auc = []
rf_precision = []
rf_recall = []
rf_f1 = []
rf_conf_matrix = np.array([[0, 0], [0, 0]])

# Loop through the folds
for train_index, test_index in kf.split(X_train_res, y_train_res):
    X_train_fold, X_val_fold = X_train_res.iloc[train_index], X_train_res.iloc[test_index]
    y_train_fold, y_val_fold = y_train_res.iloc[train_index], y_train_res.iloc[test_index]

    # Initialize and train the RandomForestClassifier
    rf = RandomForestClassifier(random_state=42)
    rf.fit(X_train_fold, y_train_fold)

    # Predict the labels
    y_pred = rf.predict(X_val_fold)
    y_pred_prob = rf.predict_proba(X_val_fold)[:, 1]

    # Append metrics for this fold
    rf_accuracy.append(accuracy_score(y_val_fold, y_pred))
    rf_roc_auc.append(roc_auc_score(y_val_fold, y_pred_prob))
    rf_precision.append(precision_score(y_val_fold, y_pred))
    rf_recall.append(recall_score(y_val_fold, y_pred))
    rf_f1.append(f1_score(y_val_fold, y_pred))

    # Update the confusion matrix
    rf_conf_matrix += confusion_matrix(y_val_fold, y_pred)

    # Plot ROC curve for each fold
    fpr, tpr, _ = roc_curve(y_val_fold, y_pred_prob)
    plt.plot(fpr, tpr, label=f'Fold ROC (AUC = {roc_auc_score(y_val_fold, y_pred_prob):.2f})')
```

```python
# Plot settings for ROC curve
plt.plot([0, 1], [0, 1], 'k--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Random Forest ROC Curve')
plt.legend(loc='best')
plt.show()

# Calculate and print average metrics
print(f'Random Forest Average Accuracy: {np.mean(rf_accuracy):.2f}')
print(f'Random Forest Average ROC AUC: {np.mean(rf_roc_auc):.2f}')
print(f'Random Forest Average Precision: {np.mean(rf_precision):.2f}')
print(f'Random Forest Average Recall: {np.mean(rf_recall):.2f}')
print(f'Random Forest Average F1-Score: {np.mean(rf_f1):.2f}')
print(f'Random Forest Confusion Matrix:\n{rf_conf_matrix}')

# Plot the Confusion Matrix
sns.heatmap(rf_conf_matrix, annot=True, fmt='d', cmap='Blues')
plt.title('Random Forest Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```

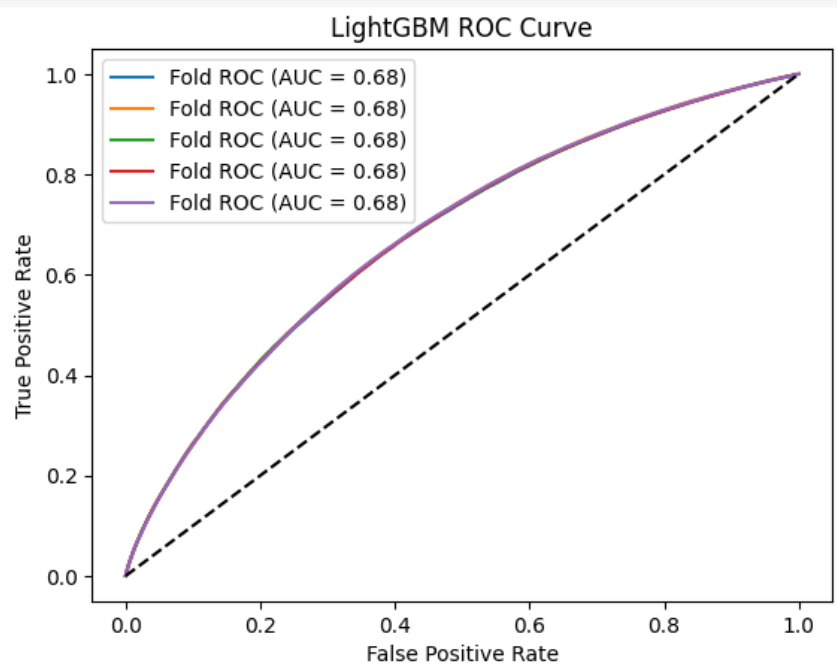



```
Random Forest Average Accuracy: 0.73
Random Forest Average ROC AUC: 0.81
Random Forest Average Precision: 0.73
Random Forest Average Recall: 0.74
Random Forest Average F1-Score: 0.73
Random Forest Confusion Matrix:
[[449060 175763]
 [160282 464541]]
```




```python
# Initialize lists to store results
lgb_accuracy = []
lgb_roc_auc = []
lgb_precision = []
lgb_recall = []
lgb_f1 = []
lgb_conf_matrix = np.array([[0, 0], [0, 0]])

# Loop through the folds
for train_index, test_index in kf.split(X_train_res, y_train_res):
    X_train_fold, X_val_fold = X_train_res.iloc[train_index], X_train_res.iloc[test_index]
    y_train_fold, y_val_fold = y_train_res.iloc[train_index], y_train_res.iloc[test_index]

    # Initialize the LGBMClassifier
    lgb = LGBMClassifier(random_state=42)

    # Train the model
    lgb.fit(X_train_fold, y_train_fold)

    # Predict the labels
    y_pred = lgb.predict(X_val_fold)
    y_pred_prob = lgb.predict_proba(X_val_fold)[:, 1]

    # Append metrics for this fold
    lgb_accuracy.append(accuracy_score(y_val_fold, y_pred))
    lgb_roc_auc.append(roc_auc_score(y_val_fold, y_pred_prob))
    lgb_precision.append(precision_score(y_val_fold, y_pred))
    lgb_recall.append(recall_score(y_val_fold, y_pred))
    lgb_f1.append(f1_score(y_val_fold, y_pred))

    # Update the confusion matrix
    lgb_conf_matrix += confusion_matrix(y_val_fold, y_pred)

    # Plot ROC curve for each fold
    fpr, tpr, _ = roc_curve(y_val_fold, y_pred_prob)
    plt.plot(fpr, tpr, label=f'Fold ROC (AUC = {roc_auc_score(y_val_fold, y_pred_prob):.2f})')

# Plot settings for ROC curve
plt.plot([0, 1], [0, 1], 'k--')
```

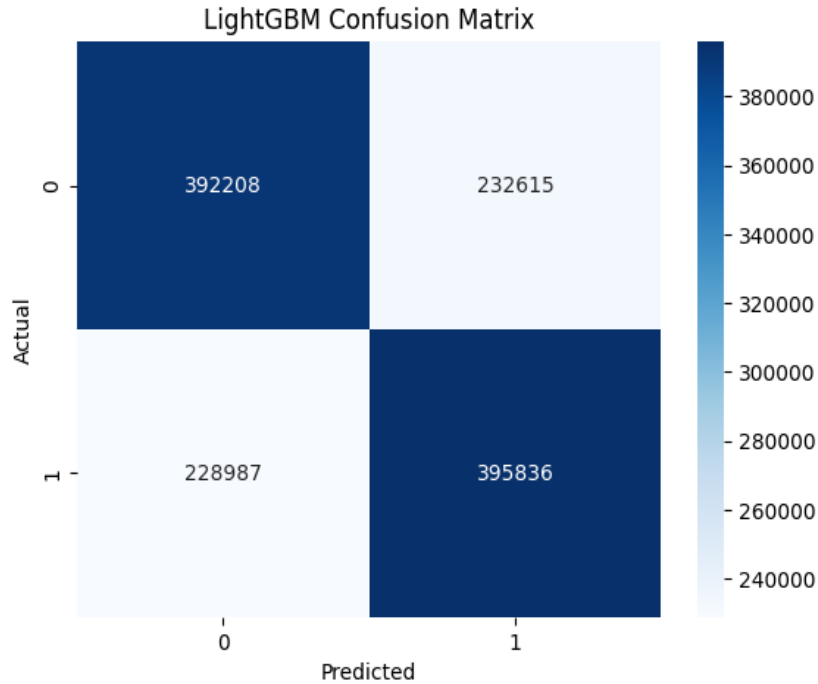
```python
# Plot settings for ROC curve
plt.plot([0, 1], [0, 1], 'k--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('LightGBM ROC Curve')
plt.legend(loc='best')
plt.show()

# Calculate and print average metrics
print(f'LightGBM Average Accuracy: {np.mean(lgb_accuracy):.2f}')
print(f'LightGBM Average ROC AUC: {np.mean(lgb_roc_auc):.2f}')
print(f'LightGBM Average Precision: {np.mean(lgb_precision):.2f}')
print(f'LightGBM Average Recall: {np.mean(lgb_recall):.2f}')
print(f'LightGBM Average F1-Score: {np.mean(lgb_f1):.2f}')
print(f'LightGBM Confusion Matrix:\n{lgb_conf_matrix}')

# Plot the Confusion Matrix
sns.heatmap(lgb_conf_matrix, annot=True, fmt='d', cmap='Blues')
plt.title('LightGBM Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```




```
LightGBM Average Accuracy: 0.63
LightGBM Average ROC AUC: 0.68
LightGBM Average Precision: 0.63
LightGBM Average Recall: 0.63
LightGBM Average F1-Score: 0.63
LightGBM Confusion Matrix:
[[392208 232615]
 [228987 395836]]
```

LightGBM Confusion Matrix

```
[228987 395836]]
```

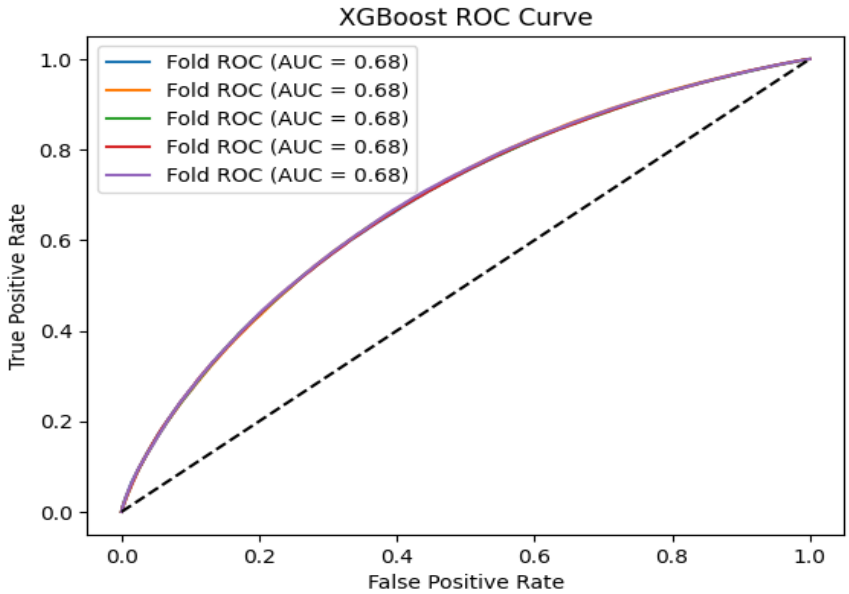



```python
# Initialize lists to store results
xgb_accuracy = []
xgb_roc_auc = []
xgb_precision = []
xgb_recall = []
xgb_f1 = []
xgb_conf_matrix = np.array([[0, 0], [0, 0]])

# Loop through the folds
for train_index, test_index in kf.split(X_train_res, y_train_res):
    X_train_fold, X_val_fold = X_train_res.iloc[train_index], X_train_res.iloc[test_index]
    y_train_fold, y_val_fold = y_train_res.iloc[train_index], y_train_res.iloc[test_index]

    # Initialize the XGBClassifier
    xgb = XGBClassifier(random_state=42, use_label_encoder=False, eval_metric='logloss')

    # Train the model
    xgb.fit(X_train_fold, y_train_fold)

    # Predict the labels
    y_pred = xgb.predict(X_val_fold)
    y_pred_prob = xgb.predict_proba(X_val_fold)[:, 1]

    # Append metrics for this fold
    xgb_accuracy.append(accuracy_score(y_val_fold, y_pred))
    xgb_roc_auc.append(roc_auc_score(y_val_fold, y_pred_prob))
    xgb_precision.append(precision_score(y_val_fold, y_pred))
    xgb_recall.append(recall_score(y_val_fold, y_pred))
    xgb_f1.append(f1_score(y_val_fold, y_pred))

    # Update the confusion matrix
    xgb_conf_matrix += confusion_matrix(y_val_fold, y_pred)

    # Plot ROC curve for each fold
    fpr, tpr, _ = roc_curve(y_val_fold, y_pred_prob)
    plt.plot(fpr, tpr, label=f'Fold ROC (AUC = {roc_auc_score(y_val_fold, y_pred_prob):.2f})')

# Plot settings for ROC curve
```

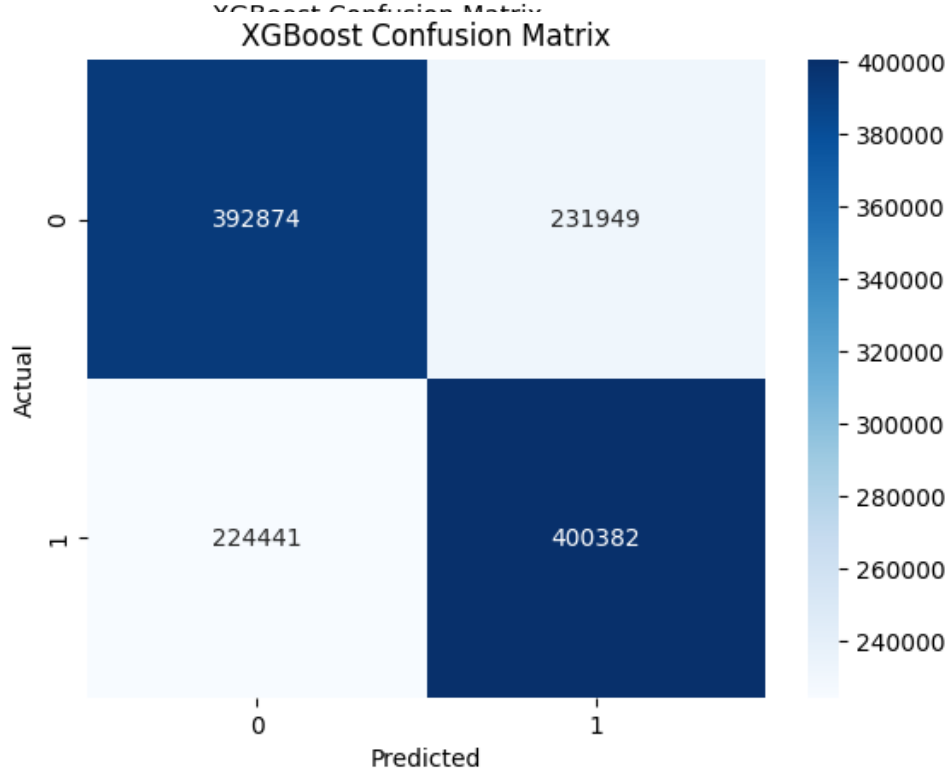
```python
# Plot settings for ROC curve
plt.plot([0, 1], [0, 1], 'k--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('XGBoost ROC Curve')
plt.legend(loc='best')
plt.show()

# Calculate and print average metrics
print(f'XGBoost Average Accuracy: {np.mean(xgb_accuracy):.2f}')
print(f'XGBoost Average ROC AUC: {np.mean(xgb_roc_auc):.2f}')
print(f'XGBoost Average Precision: {np.mean(xgb_precision):.2f}')
print(f'XGBoost Average Recall: {np.mean(xgb_recall):.2f}')
print(f'XGBoost Average F1-Score: {np.mean(xgb_f1):.2f}')
print(f'XGBoost Confusion Matrix:\n{xgb_conf_matrix}')

# Plot the Confusion Matrix
sns.heatmap(xgb_conf_matrix, annot=True, fmt='d', cmap='Blues')
plt.title('XGBoost Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```

```
/usr/local/lib/python3.10/dist-packages/xgboost/core.py:158: UserWarning: [17:46:29] WARNING: /workspace/src/learner.cc:740:
Parameters: { "use_label_encoder" } are not used.

  warnings.warn(smsg, UserWarning)
/usr/local/lib/python3.10/dist-packages/xgboost/core.py:158: UserWarning: [17:46:42] WARNING: /workspace/src/learner.cc:740:
Parameters: { "use_label_encoder" } are not used.

  warnings.warn(smsg, UserWarning)
/usr/local/lib/python3.10/dist-packages/xgboost/core.py:158: UserWarning: [17:46:54] WARNING: /workspace/src/learner.cc:740:
Parameters: { "use_label_encoder" } are not used.

  warnings.warn(smsg, UserWarning)
/usr/local/lib/python3.10/dist-packages/xgboost/core.py:158: UserWarning: [17:47:05] WARNING: /workspace/src/learner.cc:740:
Parameters: { "use_label_encoder" } are not used.

  warnings.warn(smsg, UserWarning)
```

```
warnings.warn(smsg, UserWarning)
```

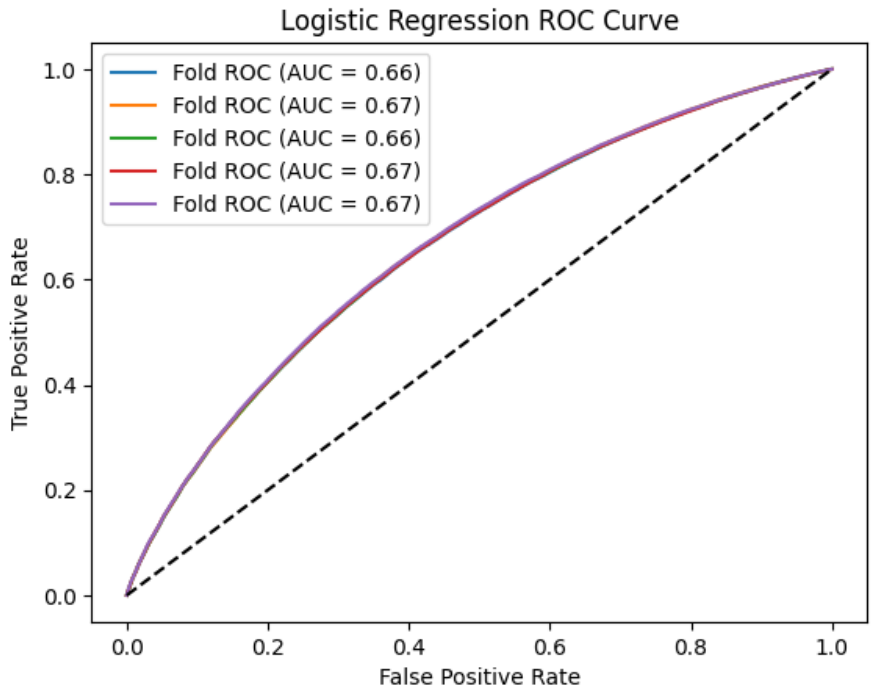



```
XGBoost Average Accuracy: 0.63
XGBoost Average ROC AUC: 0.68
XGBoost Average Precision: 0.63
XGBoost Average Recall: 0.64
XGBoost Average F1-Score: 0.64
XGBoost Confusion Matrix:
[[392874 231949]
 [224441 400382]]
```



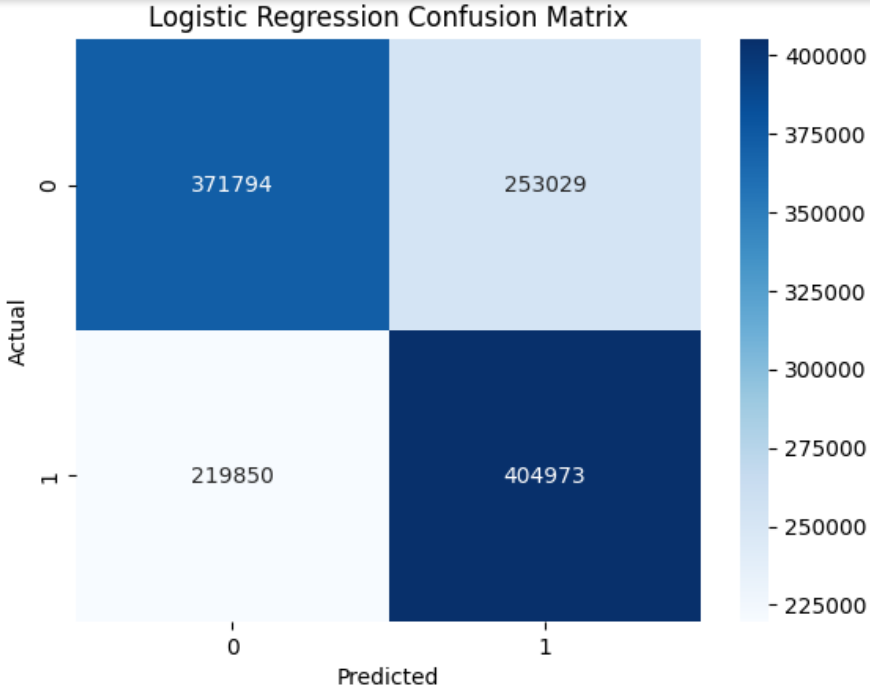
```python
# Initialize lists to store results
lr_accuracy = []
lr_roc_auc = []
lr_precision = []
lr_recall = []
lr_f1 = []
lr_conf_matrix = np.array([[0, 0], [0, 0]])

# Loop through the folds
for train_index, test_index in kf.split(X_train_res, y_train_res):
    X_train_fold, X_val_fold = X_train_res.iloc[train_index], X_train_res.iloc[test_index]
    y_train_fold, y_val_fold = y_train_res.iloc[train_index], y_train_res.iloc[test_index]

    # Initialize and train the model
    lr = LogisticRegression(random_state=42, max_iter=1000)
    lr.fit(X_train_fold, y_train_fold)

    # Predict
    y_pred = lr.predict(X_val_fold)
    y_pred_prob = lr.predict_proba(X_val_fold)[:, 1]

    # Calculate metrics
    lr_accuracy.append(accuracy_score(y_val_fold, y_pred))
    lr_roc_auc.append(roc_auc_score(y_val_fold, y_pred_prob))
    lr_precision.append(precision_score(y_val_fold, y_pred))
    lr_recall.append(recall_score(y_val_fold, y_pred))
    lr_f1.append(f1_score(y_val_fold, y_pred))

    # Confusion matrix
    lr_conf_matrix += confusion_matrix(y_val_fold, y_pred)

    # Plot ROC curve for each fold
    fpr, tpr, _ = roc_curve(y_val_fold, y_pred_prob)
    plt.plot(fpr, tpr, label=f'Fold ROC (AUC = {roc_auc_score(y_val_fold, y_pred_prob):.2f})')

# Plot settings
plt.plot([0, 1], [0, 1], 'k--')
plt.xlabel('False Positive Rate')
```

```python
# Plot settings
plt.plot([0, 1], [0, 1], 'k--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Logistic Regression ROC Curve')
plt.legend(loc='best')
plt.show()

# Output the average results
print(f'Logistic Regression Average Accuracy: {np.mean(lr_accuracy):.2f}')
print(f'Logistic Regression Average ROC AUC: {np.mean(lr_roc_auc):.2f}')
print(f'Logistic Regression Average Precision: {np.mean(lr_precision):.2f}')
print(f'Logistic Regression Average Recall: {np.mean(lr_recall):.2f}')
print(f'Logistic Regression Average F1-Score: {np.mean(lr_f1):.2f}')
print(f'Logistic Regression Confusion Matrix:\n{lr_conf_matrix}')

# Plot Confusion Matrix
sns.heatmap(lr_conf_matrix, annot=True, fmt='d', cmap='Blues')
plt.title('Logistic Regression Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```




```
Logistic Regression Average Accuracy: 0.62
Logistic Regression Average ROC AUC: 0.67
Logistic Regression Average Precision: 0.62
Logistic Regression Average Recall: 0.65
Logistic Regression Average F1-Score: 0.63
Logistic Regression Confusion Matrix:
[[371794 253029]
 [219850 404973]]
```

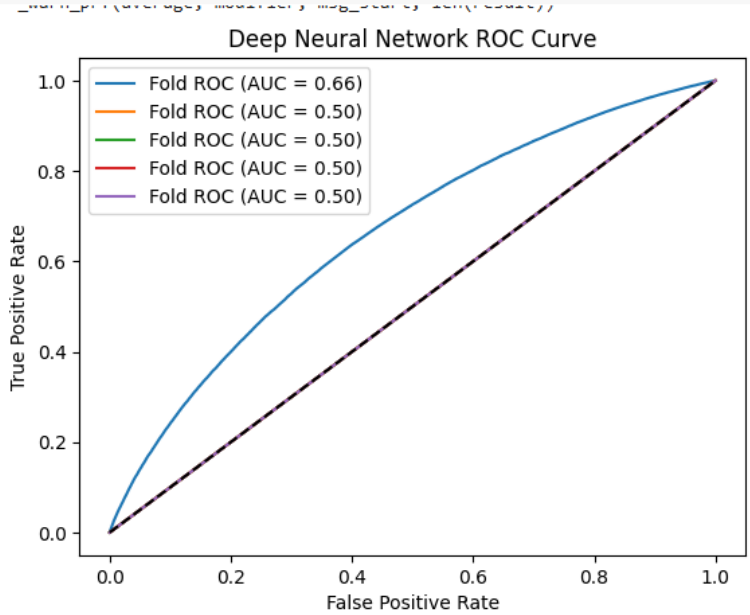
[ ]



```python
# Initialize lists to store results
dnn_accuracy = []
dnn_roc_auc = []
dnn_precision = []
dnn_recall = []
dnn_f1 = []
dnn_conf_matrix = np.array([[0, 0], [0, 0]])

# Loop through the folds
for train_index, test_index in kf.split(X_train_res, y_train_res):
    X_train_fold, X_val_fold = X_train_res.iloc[train_index], X_train_res.iloc[test_index]
    y_train_fold, y_val_fold = y_train_res.iloc[train_index], y_train_res.iloc[test_index]

    # Build the DNN model
    model = Sequential([
        Dense(64, activation='relu', input_shape=(X_train_fold.shape[1],)),
        Dropout(0.5),
        Dense(32, activation='relu'),
        Dropout(0.5),
        Dense(1, activation='sigmoid')
    ])

    # Compile the model
    model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])

    # Train the model
    model.fit(X_train_fold, y_train_fold, epochs=10, batch_size=32, verbose=0)

    # Predict
    y_pred_prob = model.predict(X_val_fold).ravel()
    y_pred = (y_pred_prob > 0.5).astype(int)

    # Calculate metrics
    dnn_accuracy.append(accuracy_score(y_val_fold, y_pred))
    dnn_roc_auc.append(roc_auc_score(y_val_fold, y_pred_prob))
    dnn_precision.append(precision_score(y_val_fold, y_pred))
    dnn_recall.append(recall_score(y_val_fold, y_pred))
    dnn_f1.append(f1_score(y_val_fold, y_pred))
```

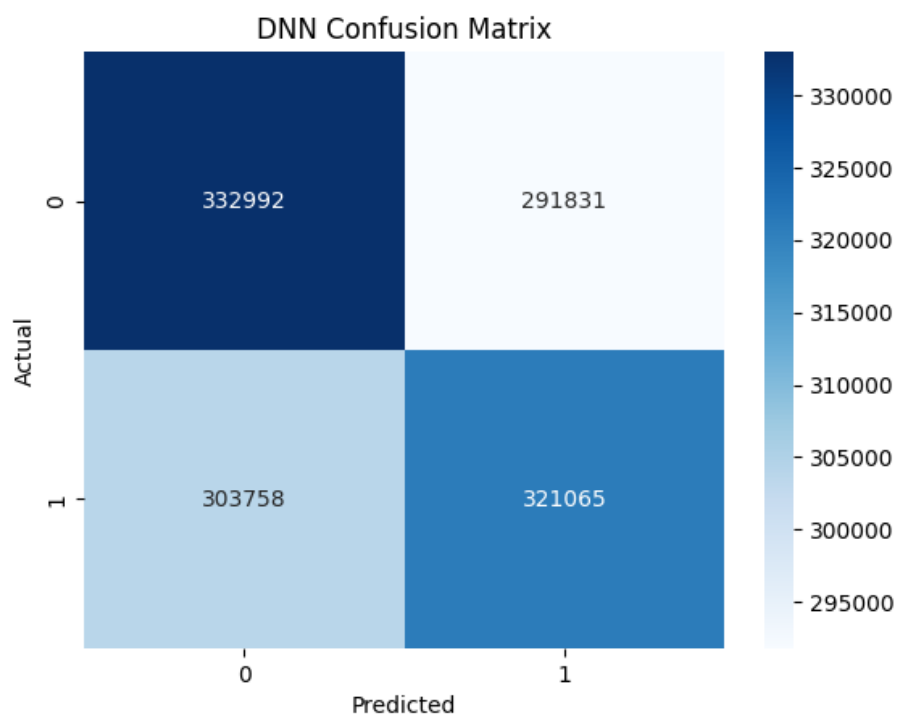
```python
    dnn_recall.append(recall_score(y_val_fold, y_pred))
    dnn_f1.append(f1_score(y_val_fold, y_pred))

    # Confusion matrix
    dnn_conf_matrix += confusion_matrix(y_val_fold, y_pred)

    # Plot ROC curve for each fold
    fpr, tpr, _ = roc_curve(y_val_fold, y_pred_prob)
    plt.plot(fpr, tpr, label=f'Fold ROC (AUC = {roc_auc_score(y_val_fold, y_pred_prob):.2f})')

# Plot settings
plt.plot([0, 1], [0, 1], 'k--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Deep Neural Network ROC Curve')
plt.legend(loc='best')
plt.show()

# Output the average results
print(f'DNN Average Accuracy: {np.mean(dnn_accuracy):.2f}')
print(f'DNN Average ROC AUC: {np.mean(dnn_roc_auc):.2f}')
print(f'DNN Average Precision: {np.mean(dnn_precision):.2f}')
print(f'DNN Average Recall: {np.mean(dnn_recall):.2f}')
print(f'DNN Average F1-Score: {np.mean(dnn_f1):.2f}')
print(f'DNN Confusion Matrix:\n{dnn_conf_matrix}')

# Plot Confusion Matrix
sns.heatmap(dnn_conf_matrix, annot=True, fmt='d', cmap='Blues')
plt.title('DNN Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```

```
/usr/local/lib/python3.10/dist-packages/keras/src/layers/core/dense.py:87: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When usin
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
7811/7811 ━━━━━━━━━━━━━━━━━━━━ 10s 1ms/step
/usr/local/lib/python3.10/dist-packages/keras/src/layers/core/dense.py:87: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When usin
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
7811/7811 ━━━━━━━━━━━━━━━━━━━━ 9s 1ms/step
```




```
DNN Average Accuracy: 0.52
DNN Average ROC AUC: 0.53
DNN Average Precision: 0.33
DNN Average Recall: 0.51
DNN Average F1-Score: 0.39
DNN Confusion Matrix:
[[332992 291831]
 [303758 321065]]
```

```
[303758 321065]]
```

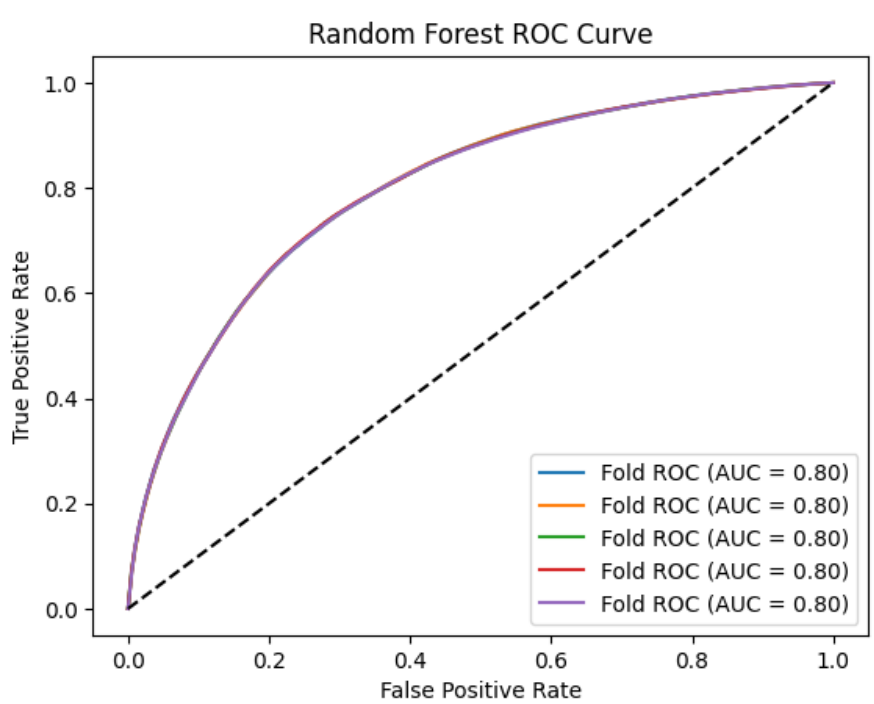


```python
# Import necessary libraries
from imblearn.over_sampling import SMOTE
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# Define your features (X) and target variable (y)
X = data[[ 'age', 'gender','cancer_stage', 'bmi', 'cholesterol_level', 'hypertension', 'asthma', 'cirrhosis', 'other_cancer', ]]
y = data['survived']

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Apply SMOTE to the training data
smote = SMOTE(random_state=42)
X_train_res, y_train_res = smote.fit_resample(X_train, y_train)


# Check the distribution of the resampled training data
print("Training set after SMOTE:")
print(y_train_res.value_counts())

# Check the distribution of the original testing data
print("\nTesting set distribution:")
print(y_test.value_counts())
```

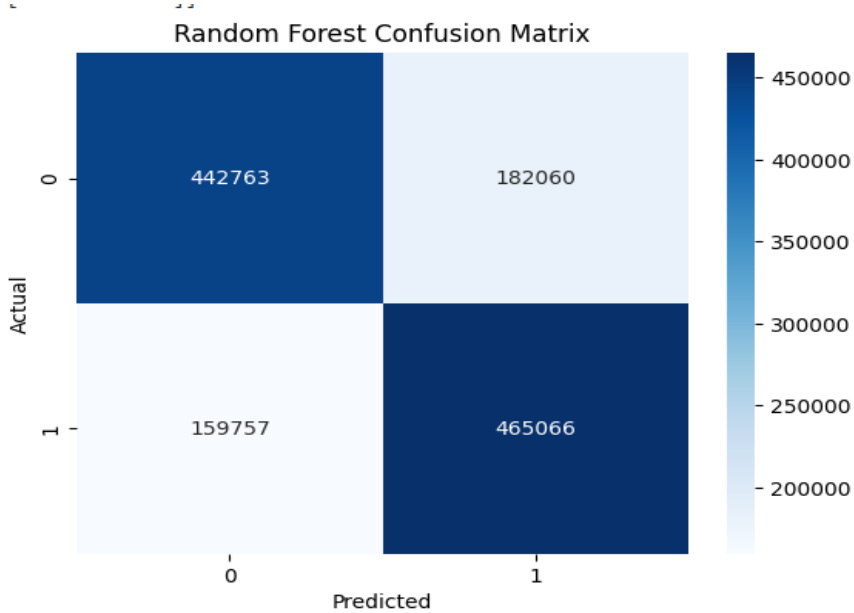
```
Training set after SMOTE:
survived
1    624823
0    624823
Name: count, dtype: int64

Testing set distribution:
survived
0    155718
1     44282
Name: count, dtype: int64
```

```python
# Initialize StratifiedKFold
kf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

# Initialize lists to store results
rf_accuracy = []
rf_roc_auc = []
rf_precision = []
rf_recall = []
rf_f1 = []
rf_conf_matrix = np.array([[0, 0], [0, 0]])

# Loop through the folds
for train_index, test_index in kf.split(X_train_res, y_train_res):
    X_train_fold, X_val_fold = X_train_res.iloc[train_index], X_train_res.iloc[test_index]
    y_train_fold, y_val_fold = y_train_res.iloc[train_index], y_train_res.iloc[test_index]

    # Initialize and train the RandomForestClassifier
    rf = RandomForestClassifier(random_state=42)
    rf.fit(X_train_fold, y_train_fold)

    # Predict the labels
    y_pred = rf.predict(X_val_fold)
    y_pred_prob = rf.predict_proba(X_val_fold)[:, 1]

    # Append metrics for this fold
    rf_accuracy.append(accuracy_score(y_val_fold, y_pred))
    rf_roc_auc.append(roc_auc_score(y_val_fold, y_pred_prob))
    rf_precision.append(precision_score(y_val_fold, y_pred))
    rf_recall.append(recall_score(y_val_fold, y_pred))
    rf_f1.append(f1_score(y_val_fold, y_pred))

    # Update the confusion matrix
    rf_conf_matrix += confusion_matrix(y_val_fold, y_pred)

    # Plot ROC curve for each fold
    fpr, tpr, _ = roc_curve(y_val_fold, y_pred_prob)
    plt.plot(fpr, tpr, label=f'Fold ROC (AUC = {roc_auc_score(y_val_fold, y_pred_prob):.2f})')
```

```python
# Plot settings for ROC curve
plt.plot([0, 1], [0, 1], 'k--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Random Forest ROC Curve')
plt.legend(loc='best')
plt.show()

# Calculate and print average metrics
print(f'Random Forest Average Accuracy: {np.mean(rf_accuracy):.2f}')
print(f'Random Forest Average ROC AUC: {np.mean(rf_roc_auc):.2f}')
print(f'Random Forest Average Precision: {np.mean(rf_precision):.2f}')
print(f'Random Forest Average Recall: {np.mean(rf_recall):.2f}')
print(f'Random Forest Average F1-Score: {np.mean(rf_f1):.2f}')
print(f'Random Forest Confusion Matrix:\n{rf_conf_matrix}')

# Plot the Confusion Matrix
sns.heatmap(rf_conf_matrix, annot=True, fmt='d', cmap='Blues')
plt.title('Random Forest Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```




```
Random Forest Average Accuracy: 0.73
Random Forest Average ROC AUC: 0.80
Random Forest Average Precision: 0.72
Random Forest Average Recall: 0.74
Random Forest Average F1-Score: 0.73
Random Forest Confusion Matrix:
[[442763 182060]
 [159757 465066]]
```

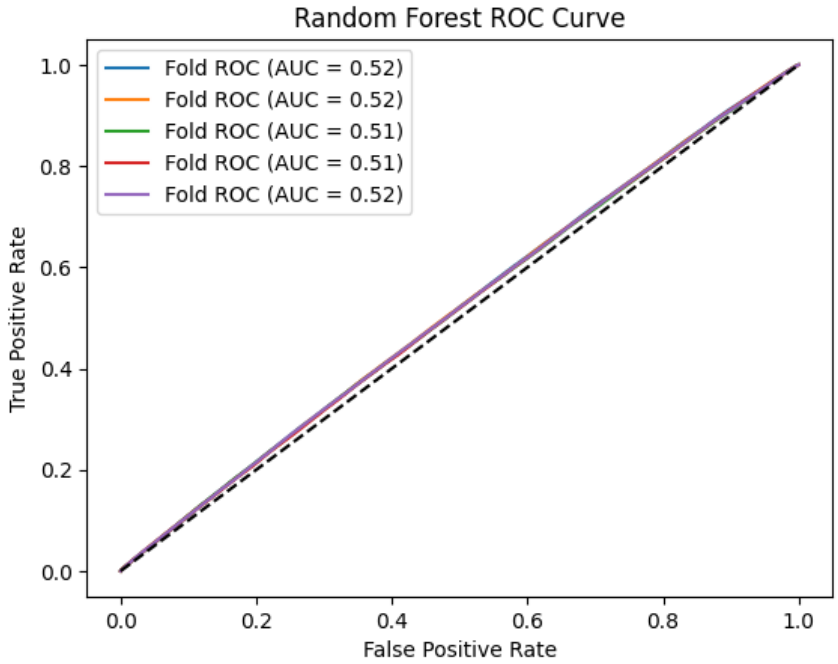


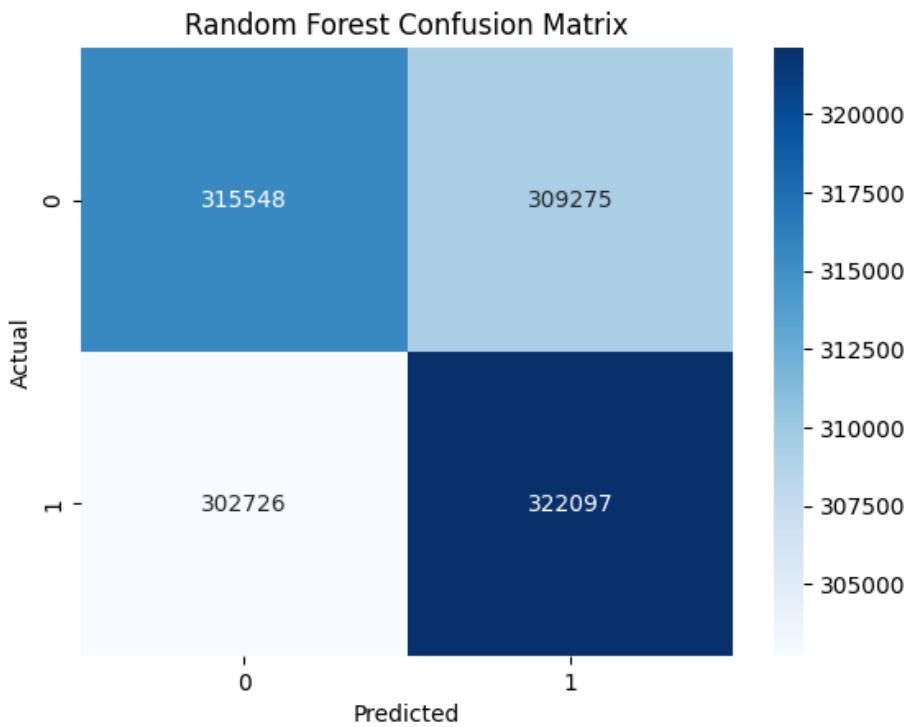
```python
# Import necessary libraries
from imblearn.over_sampling import SMOTE
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# Define your features (X) and target variable (y)
X = data[['treatment_type', 'family_history', 'smoking_status']]
y = data['survived']

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Apply SMOTE to the training data
smote = SMOTE(random_state=42)
X_train_res, y_train_res = smote.fit_resample(X_train, y_train)


# Check the distribution of the resampled training data
print("Training set after SMOTE:")
print(y_train_res.value_counts())

# Check the distribution of the original testing data
print("\nTesting set distribution:")
print(y_test.value_counts())
```

```
Training set after SMOTE:
survived
1    624823
0    624823
Name: count, dtype: int64

Testing set distribution:
survived
0    155718
1     44282
Name: count, dtype: int64
```

```python
# Initialize StratifiedKFold
kf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

# Initialize lists to store results
rf_accuracy = []
rf_roc_auc = []
rf_precision = []
rf_recall = []
rf_f1 = []
rf_conf_matrix = np.array([[0, 0], [0, 0]])

# Loop through the folds
for train_index, test_index in kf.split(X_train_res, y_train_res):
    X_train_fold, X_val_fold = X_train_res.iloc[train_index], X_train_res.iloc[test_index]
    y_train_fold, y_val_fold = y_train_res.iloc[train_index], y_train_res.iloc[test_index]

    # Initialize and train the RandomForestClassifier
    rf = RandomForestClassifier(random_state=42)
    rf.fit(X_train_fold, y_train_fold)

    # Predict the labels
    y_pred = rf.predict(X_val_fold)
    y_pred_prob = rf.predict_proba(X_val_fold)[:, 1]

    # Append metrics for this fold
    rf_accuracy.append(accuracy_score(y_val_fold, y_pred))
    rf_roc_auc.append(roc_auc_score(y_val_fold, y_pred_prob))
    rf_precision.append(precision_score(y_val_fold, y_pred))
    rf_recall.append(recall_score(y_val_fold, y_pred))
    rf_f1.append(f1_score(y_val_fold, y_pred))

    # Update the confusion matrix
    rf_conf_matrix += confusion_matrix(y_val_fold, y_pred)

    # Plot ROC curve for each fold
    fpr, tpr, _ = roc_curve(y_val_fold, y_pred_prob)
    plt.plot(fpr, tpr, label=f'Fold ROC (AUC = {roc_auc_score(y_val_fold, y_pred_prob):.2f})')
```

```python
# Plot settings for ROC curve
plt.plot([0, 1], [0, 1], 'k--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Random Forest ROC Curve')
plt.legend(loc='best')
plt.show()

# Calculate and print average metrics
print(f'Random Forest Average Accuracy: {np.mean(rf_accuracy):.2f}')
print(f'Random Forest Average ROC AUC: {np.mean(rf_roc_auc):.2f}')
print(f'Random Forest Average Precision: {np.mean(rf_precision):.2f}')
print(f'Random Forest Average Recall: {np.mean(rf_recall):.2f}')
print(f'Random Forest Average F1-Score: {np.mean(rf_f1):.2f}')
print(f'Random Forest Confusion Matrix:\n{rf_conf_matrix}')

# Plot the Confusion Matrix
sns.heatmap(rf_conf_matrix, annot=True, fmt='d', cmap='Blues')
plt.title('Random Forest Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```




```
Random Forest Average Accuracy: 0.51
Random Forest Average ROC AUC: 0.52
Random Forest Average Precision: 0.51
Random Forest Average Recall: 0.52
Random Forest Average F1-Score: 0.51
Random Forest Confusion Matrix:
[[315548 309275]
 [302726 322097]]
```